World Health Organization

European Region

Air quality indexes

Key considerations and roadmaps for best practices

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Abstract

Air quality indexes (AQIs) are widely used to communicate short-term air pollution concentrations and related health risks to the public. Conventional AQIs are typically formulated based on the concentration of the single air pollutant that most exceeds its regulatory standard (among all pollutants measured). Alternatively, health-based AQIs represent the combined health risks due to multiple air pollutants and are formulated from concentration–response functions derived from epidemiological evidence. This report examines public health approaches to improve AQIs, with a focus on roadmaps for best practices in developing, validating and communicating health-based AQIs. It first reviews the status of and differences between conventional and health-based AQIs. The Canadian Air Quality Health Index is presented as a model health-based approach, and the strengths and weaknesses of several novel indexes are discussed. Next, published studies supporting conventional and health-based AQIs across several tiers of evaluation are reviewed, focusing on evidence of public health benefits. The roles of public communication and global equity are also discussed. From this assessment, key considerations that serve as the basis for framing roadmaps for best practices moving forward are identified. In conclusion, health-based AQIs are noted to offer several advantages compared with conventional AQIs. There is a need to adopt locally-adapted, equity-sensitive approaches that reflect diverse air pollution profiles, health susceptibilities and cultural contexts. Strengthening risk communication – through improved indexes and innovative strategies – plays a key role in supporting future efforts to protect public health.

Keywords

AIR POLLUTION, AIR POLLUTANTS, ENVIRONMENTAL EXPOSURE, ENVIRONMENT AND PUBLIC HEALTH, HEALTH COMMUNICATION, PRIMARY PREVENTION

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Contents

Acknowledgements ii

Abbreviations iii

1. Introduction 1

1.1 Background 1

1.2 Rationale and objectives 3

1.3 Methodology 4

2. Scientific review 5

2.1 Overview of AQIs 5

2.2 Existing and novel health-based AQIs 9

2.3 Evaluation of AQIs and alerts 16

2.4 Public communication of AQIs 27

2.5 Global equity 31

3. Key considerations and roadmaps for future AQ(H)Is 33

3.1 Development 33

3.2 Communication 39

3.3 Validation 40

3.4 Further refinement and global equity 42

4. Conclusions 45

References 47

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Abbreviations

AQG	air quality guideline (levels)
AQHI	air quality health index
AQI	air quality index
CO	carbon monoxide
CVD	cardiovascular disease
EU	European Union
LCS	low-cost sensors
O_3	ozone
NO_2	nitrogen dioxide
$PM_{2.5}$	particulate matter of ≤ 2.5 μm in aerodynamic diameter
PM_{10}	particulate matter of ≤ 10 μm in aerodynamic diameter
SO_2	sulfur dioxide

1. Introduction

1.1 Background

WHO estimates that globally 4.2 million deaths per year and a loss of 105.2 million disability-adjusted life years are attributable to ambient (outdoor) air pollution (principally derived from fossil fuel combustion) *(1,2)*. These health figures reflect the impact of long-term exposure to fine particulate matter of ≤ 2.5 µm in aerodynamic diameter ($PM_{2.5}$) on five diseases: acute lower respiratory infections, chronic obstructive pulmonary disease, ischaemic heart disease, stroke and lung cancer. As additional diseases (e.g. type 2 diabetes, neurological disorders) and pollutants (e.g. nitrogen dioxide (NO_2) and ozone (O_3)) are not considered, the total global public health burden from all ambient air pollution is likely to be even larger *(3,4)*.

In 2021 WHO reduced most of its air quality guideline (AQG) levels in accordance with accruing data that show adverse health effects at low air pollutant concentrations with no observable thresholds below which long-term exposures can be considered safe, especially for $PM_{2.5}$ *(5)*. Under these updated criteria, 99% of the global population live in areas where air pollution concentrations exceed annual WHO AQG levels *(6,7)*. However, ambient air pollution levels vary substantially worldwide. Low- and middle-income countries face $PM_{2.5}$ levels that are roughly threefold higher than in high-income countries *(5,7,8)*. The WHO African, South-East Asia and Eastern Mediterranean regions are those most heavily impacted by poor air quality. While lowering trends in air pollution concentrations have been reported in many (but not all) locations, there is rising concern that climate change may hamper further improvements or even worsen air quality in the future *(9)*.

In addition to the immense public health threat posed by long-term exposure to air pollution, robust evidence shows that even short-term elevations (e.g. for 24 hours) in air pollution levels promote acute increases in morbidity and mortality independent of (and for all levels of) the chronic prevailing concentrations *(10–12)*. Approximately 1 million deaths worldwide are attributable each year to acute (i.e. 24-hour) elevations in $PM_{2.5}$ levels alone *(12)*. In this context, government agencies in many countries have developed and regularly issue air quality indexes (AQIs) (Box 1) *(13)*. Their aim is to communicate easy-to-understand information to the public on local short-term air quality, as well as the related health risks *(14–16)*. Practical advice on behaviour changes for individuals is also often provided to help to reduce exposures.

Box 1. Report terminology

Air pollution indexes

AQI: conventional indexes calculated from short-term individual air pollution concentrations with thresholds most commonly derived from national regulations. In general, the single air pollutant (among all pollutants measured) that most exceeds its regulatory standard sets the AQI level.

Health-based AQI/air quality health index (AQHI): indexes of this type represent short-term health risks from (typically) multipollutant air pollution exposures derived from epidemiological evidence. This report uses the term AQHI synonymously with health-based AQI, except when referring specifically to the Canadian AQHI.

Air quality alert: the active issuance of warnings to the public (e.g. via the internet, text messages on mobile networks) when AQIs, AQHIs or individual air pollution concentrations reach specified high thresholds.

AQ(H)Is: encompasses both conventional AQIs and AQHIs.

Raw air quality data: individual air pollutant concentrations, as released to the public.

AQ(H)I values: numbers (i.e. unitless values) denoting the concentration of a specific air pollutant across a uniform scale (AQI) or the risk of a specific health outcome (AQHI).

AQ(H)I categories or bands: a range of AQ(H)I values comprising air pollution exposures (AQIs) or health risks (AQHIs) for which the same behavioural advice is issued for subsets of the population.

Geographical scales of indexes

Local: single communities or cities.
Regional: counties, states, provinces and governmental regions.
National: individual countries.
Multicountry: a group of neighbouring countries, for example, the European Union (EU).

Populations

Susceptible: individuals/populations who are prone to greater health risks than healthy adults at the same concentration of pollution exposures.

Vulnerable: individuals/populations who face more frequent and/or higher exposures to air pollution.

Understanding and applying information

Health literacy: the personal knowledge and competencies that accumulate through daily activities, social interactions and across generations. These are mediated by the organizational structures and availability of resources that enable people to access, understand, appraise and use information and services in ways that promote and maintain good health and well-being for themselves and those around them *(17)*.

Environmental health literacy: the basic knowledge and skills needed to comprehend environmental health risks and to devise, assess, implement and evaluate potential solutions *(18)*. The level of environmental health literacy determines how individuals and communities make sense of and act on health-related information about environmental hazards *(18)*.

1.2 Rationale and objectives

Effective communication on the health risks associated with air pollution is an important element of several WHO policy frameworks. World Health Assembly resolution WHA68.8 on health and the environment: addressing the health impacts from air pollution *(19)* and the accompanying roadmap *(20)* call for action to inform policy-makers and the public about the adverse health effects of air pollution and the benefits of implementing appropriate actions. Similarly, the *WHO global strategy on health, environment and climate change* identified the critical importance of enhanced communication, increased awareness and active public engagement *(21)*.

At European level, in 2017 the Ostrava Declaration of the Sixth Ministerial Conference on Environment and Health recognized the responsibility of authorities to improve public understanding of air quality issues through targeted communication strategies *(22)*. Building on this, in 2023 the Budapest Declaration of the Seventh Ministerial Conference advocated strengthening communication tools and improving literacy regarding the interconnectedness of health, environment and climate change *(23)*. This Declaration also highlighted the importance of expanding access to information and facilitating public participation in decision-making processes related to the environment and health.

In May 2025 the Seventy-eighth World Health Assembly adopted the *Updated road map for an enhanced global response to the adverse health effects of air pollution (24)*, which paves the way for global health action until 2030. The roadmap identified strategic communication as an important component to facilitate the integration of air pollution reduction efforts within broader health and development initiatives. Recognizing that community awareness and the dissemination of accurate, evidence-based information are fundamental to catalysing multisectoral action, the roadmap identified the health sector as a key agent in disseminating evidence on health risks and exposure reduction strategies.

At global level, the promotion of and support for the right of access for all to information about pollution and pollutants is one of the intended outcomes of the joint efforts of *The United Nations system: common approach towards a pollution-free planet (25)*.

At EU level, Directive (EU) 2024/2881 on ambient air quality and cleaner air for Europe should be transposed into national laws by 2026. The Directive stipulates several requirements related to the provision of air quality information to the public. Each EU Member State is required to make publicly available a clear and easily understandable AQI that provides hourly updates for key pollutants. Wherever feasible, this AQI should be comparable across countries, follow WHO recommendations and be based on the European Air Quality Index *(26,27)*, including health-related information for at-risk groups. Countries may use the European AQI directly (or develop their own), but are requested to reference it if they choose not to adopt it.

A generic communication requirement involves sharing information on symptoms linked to high air pollution levels and on reducing personal exposure, especially in areas frequented by at-risk populations, such as health-care facilities *(28)*.

In response to a call by experts for WHO to promote health-based AQIs *(29)* and building on WHO's recent work to support evidence-based risk communication *(5,15,29–31)*, including at the country level *(32)*, the current report aims to provide a comprehensive overview of AQIs from a public health perspective. The report focuses on health-based AQIs or AQHIs, including a review of existing practices, salient issues and key considerations to strengthen their implementation. The main target audiences are authorities in charge of providing air quality information to the public, as well as technical experts, health-care professionals and researchers in the field of air quality.

1.3 Methodology

A structured approach was applied to guide the development of this report, combining a targeted review of the literature, critical appraisal by an expert writing group, and an external review process. The expert writing group had expertise in atmospheric science, clinical medicine, communication, environmental epidemiology, health policy, population health and toxicology. The evidence base was compiled through a PubMed search for peer-reviewed literature conducted in October 2024 using relevant search terms, supplemented by grey literature, including WHO reports and other key documents identified by the expert writing group. Relevant papers were summarized to capture current practices and key considerations, with the aim of informing a roadmap for best practices. Insights from the expert writing group were incorporated throughout the process through structured email exchanges and virtual meetings. A preliminary draft of the report underwent external review by experts and stakeholders, who contributed additional perspectives in areas such as environmental engineering and behavioural science. Their feedback was assessed and incorporated where appropriate.

Throughout the report, the Canadian AQHI, the United Kingdom's daily AQI and the United States AQI were often highlighted as illustrative examples because they were (among) the first official indexes in their respective categories, have long records of real-world use and have more extensively informed research and development efforts.

The report is organized into three further chapters: Chapter 2 provides a review of the science related to AQ(H)Is, Chapter 3 contains key considerations and roadmaps to enhance current practices, and Chapter 4 contains concluding remarks.

2. Scientific review

2.1 Overview of AQIs

2.1.1 What is an AQI?

The threat of ambient air pollution to human health *(2,6,7)* and evidence that reduced exposures yield observable health benefits *(33,34)* call for the provision of accessible and useful information to protect the public from poor air quality *(5,29,30)*. For short-term fluctuations in air quality, one mechanism for achieving this includes AQIs. Official AQIs are regulatory-based communication and risk-awareness tools issued by government agencies in many countries on a frequent basis *(15,35)*. Around the world, AQIs vary in relation to many factors, including thresholds (or breakpoints) for categories (or bands) owing to prevailing pollution concentrations, number of AQI categories and their pollution ranges, formulations for calculations, whether they are derived from single pollutants or mixtures, relevant geographical areas, and whether they are considered health based, such as the Canadian AQHI (Box 1) *(13,16)*. As reviewed in a 2023 WHO Regional Office for Europe report, country-specific AQIs and public communication strategies differ substantially within the WHO European Region *(15)*. All AQI types, however, communicate information to the public only about short-term local air quality and related health risks with the intention of influencing people's behaviour (over hours to days) and reducing pollution-related acute health risks *(36)*.

2.1.2 Construction of AQIs

To be informative for the public, AQIs must translate complex air quality information into an understandable, clear and concise format. This is achieved by converting air pollution concentration data (i.e. physical units) from fixed regulatory monitoring stations and modelled forecasts into an accessible representation of air quality conditions (often, unitless values) and health effects more likely to be experienced on the day described by the AQI or soon afterwards (i.e. the short term) *(37)*. The rationale for converting different pollutant levels (e.g. NO_2, O_3, $PM_{2.5}$) into standardized AQI values is to reduce complexity because individual air pollutants vary in concentration ranges, measurement units (e.g. µg/m^3, ppb) and concentration–response relationships. This process allows for easier comparison of the relative concentrations (and health risks) of the various pollutants using a uniform scale (e.g. ranging from 0 to 500). The mathematical formulae that are used to convert an air pollutant concentration into a unitless AQI numerical value differ around the world and are updated in accordance with changes to regulations. They generally include a linear scale (e.g. in Australia, where the AQI is calculated in direct proportion to the air quality standards), normalized or nonlinear scale (e.g. the United States AQI, where

concentrations of each pollutant are converted into a 0–500 scale) and category (band) system (e.g. in the United Kingdom, where a 10-point scale is divided into low, moderate, high and very high bands) *(35)*.

Formulation of an AQI is usually based on the national air quality standards for each country. These standards are formulated based on the short-term health effects of the index pollutants, current levels of pollutants, scientific evidence emerging from toxicological and epidemiological research, developments in national limit values, and air quality objectives *(37)*. A specific AQI value (e.g. 100 in the United States of America) is typically set at the ambient concentration for each pollutant that corresponds to the short-term National Ambient Air Quality Standard. The commonly selected air pollutants that are used to calculate AQIs include those that are currently known to have the greatest short-term health effects on the local population, namely NO_2, O_3 and $PM_{2.5}$ (or particulate matter of ≤ 10 µm in aerodynamic diameter (PM_{10})). Some countries (e.g. United States) also incorporate carbon monoxide (CO) and sulfur dioxide (SO_2).

Each pollutant is compared with a benchmark that represents the ideal state of air quality. At national level, individual countries determine their benchmark by accounting for factors such as level of economic development and air pollution control capacity. Most commonly, the national air quality standards of each jurisdiction are used as the benchmark for AQI construction *(35)*. The pollutant with the highest value relative to its limit value is typically reported as the overall AQI value for the designated time. For example, if the forecast or measurement is Moderate for O_3 and Low for PM_{10}, then the overall value assigned will be Moderate. Disregarding other pollutants with smaller or equivalent values means that potential cumulative or synergistic effects of multiple air pollutants cannot be captured, even though they may exist and most pollution episodes are not confined to a single pollutant *(38)*.

Averaging times to calculate AQIs often differ between pollutants (e.g. 8 hours for O_3, 24 hours for $PM_{2.5}$), reflecting the exposure windows used in many epidemiological studies that show acute health effects *(10–12)*. As AQI values for O_3 and $PM_{2.5}$ relate to relatively long averaging times, delays can occur between the onset of air pollution episodes, an AQI reaching a warning level and the issuance of behavioural advice to limit exposure. For example, the official governmental AQI for $PM_{2.5}$ could not technically be reported until after the end of the day (i.e. the 24-hour averaging period is complete). To overcome this shortcoming and inform the public about air quality in approximately real time, rapidly changing AQIs are often reported (e.g. on nongovernmental internet websites). These represent AQIs calculated from pollution levels averaged (or modelled) over briefer periods, such as the last hour, or weighted towards the most recent few hours *(39)*. Forecasted AQI values for the following hours or days are also often reported *(40)*.

An important contextual point that can be taken from such averaging times is that all AQIs in their current form are solely intended to inform and help to protect the public from relatively high levels of air pollution (i.e. concentrations above the AQI thresholds set for a country or region) that occur intermittently over the short term (i.e. hours to a day). They do not aim to inform the public about or mitigate the adverse health effects of long-term exposure to lower concentrations of air pollutants, which account for a greater public health burden and must be addressed by other means (e.g. annual AQG levels).

AQIs consist of several bands or categories, sometimes named and often differing across countries, that indicate either the air quality (e.g. good or poor) or air pollution level (e.g. low or high). The level of risk to

human health can be inferred because it is monotonically associated with pollution level and, thus, AQI value. The bands may be further divided into a numeric scale representing the calculated AQI value, with the format and presentation varying by country. The higher the band/number, the greater the level of air pollution, health risk and need to take precautions. These scales often incorporate colour coding to aid interpretation and use equivalent colours on maps posted on AQI websites and apps to allow users to quickly determine the local air quality *(41)*. The setting of AQI breakpoints between bands is mainly based on (inter)national air pollution standards or guidelines and is largely an arbitrary process *(35,42)*. Indeed, epidemiological studies have shown that, at population level, no thresholds of effect can be identified for the common air pollutants; thus, an impact can be expected for some individuals even at low levels of exposure *(43–45)*.

2.1.3 Health-based AQIs

Limitations of single-pollutant-oriented (conventional) AQIs include having (i) the potential to poorly communicate non-threshold relationships between short-term air pollution exposures and health effects (especially when AQI categories rather than individual values are considered); (ii) an inability to capture the cumulative health risks of multiple air pollutants; and (iii) a link to regulatory standards that might be influenced by factors other than health risk. These issues have led some countries to develop health-based multipollutant indexes, or AQHIs *(46–49)*, as follows. First, various health risks related to individual pollutants are calculated based on formulae constructed using the full range of concentration–response functions from local epidemiological analyses (e.g. time-series studies) *(46)*. Next, the risks associated with each pollutant are combined. The specific health outcome (e.g. mortality), air pollutants and mathematical methods to combine health risks vary among AQHIs. As such, AQHIs comprehensively reflect a specific short-term health risk (e.g. daily mortality, emergency department visits for acute ischaemic stroke or asthma) for different mixtures of air pollutants. They also capture the significant gradation of health risks across low levels of exposures to multiple pollutants, even those falling below regulatory standards – which conventional AQIs fail to do *(35)*. The AQHI values are then adjusted to a scale, further grouped into categories to reflect different levels of health risks and reported for both the current air quality situation and forecasted to provide warnings for subsequent days. In summary, AQIs provide a scale related to the concentration of the pollutant whose level most exceeds its national standard, whereas AQHIs provide a scale related to the short-term health risks of exposure to a mixture of pollutants (Fig. 1).

AQ(H)Is may serve as counselling tools for the public by providing behavioural guidance linked to the index categories. Higher AQ(H)Is (i.e. worse air quality) are stepwise linked to more intense/stringent advice for behaviour changes. Typically, susceptible groups (Box 1) are advised to make behaviour changes at lower index categories *(16)*. Examples include people with asthma, pulmonary disease and cardiovascular disease (CVD); pregnant women; older people; and children *(16)*. Within each index category, specific health messages are tailored to the risk of adverse health effects among various populations to limit acute exposures and mitigate potential health effects. Advice during periods of poor air quality may be to reduce or reschedule outdoor activities, use public transport to reduce the number of private vehicles on the road or check that medication is being taken as advised *(15)*. Vulnerable populations (Box 1) may also be considered during AQI development. Examples include people living in urban areas or in proximity to fossil fuel emission sources (e.g. main roads, industrial activities), people with a lower socioeconomic

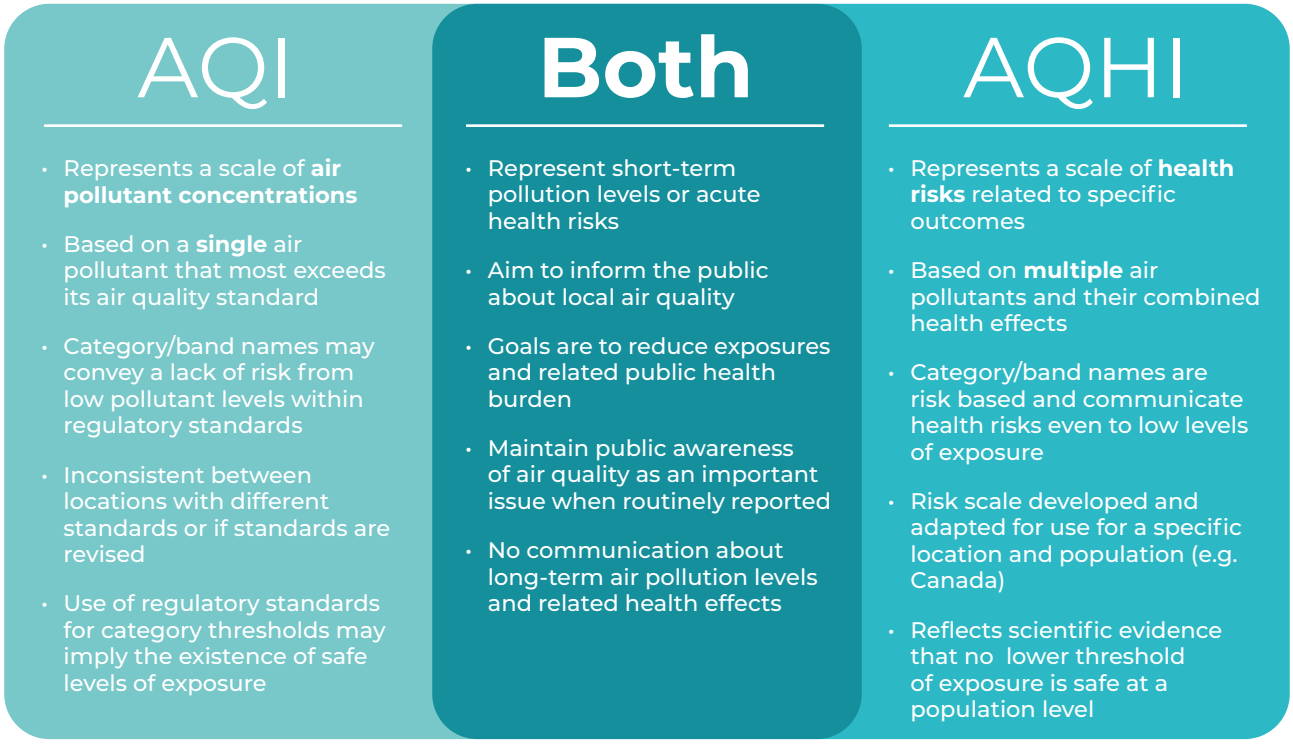
status and racial/ethnic minority populations (often, but not exclusively), and many citizens of low- to middle-income countries globally. Elevated AQIs can also serve as triggers for authorities to implement interventions such as traffic restrictions, free/reduced fees for public transportation, restrictions on solid fuel use for heating and limitations on high industrial emitters.

2.1.4 Implications of the differences between AQIs and AQHIs

By capturing health risks due to the combined actions of multiple air pollutants (not only the single most elevated pollutant), AQHI values convey more information than AQIs to the public. This is especially relevant when some or all pollutant levels are low (below regulatory standards). AQHIs correlate more strongly than AQIs with health risks (reviewed in sections 2.2.2 and 2.3). Increases along the AQHI scale and the threshold values chosen for categories represent uniform stepwise elevations in health risks derived from epidemiological data rather than the irregular thresholds of pollution levels set by regulations for AQIs that are of no direct health relevance. Individual AQI values (e.g. 1–500) represent converted air pollutant concentrations (section 2.1.2.). Therefore, the full range of AQI values conveys indirect (and perhaps overly precise) estimations of health risks on a continuous scale. As such, adverse health risks at low exposures below thresholds derived from guidelines or regulatory standards (e.g. AQI values of 1–49 in the United States) are indirectly conveyed by conventional AQIs. However, it is highly probable (albeit not well studied) that the public pays more attention to the colour-coded AQI categories (or bands) than to exact AQI values, especially since health messages and behavioural guidance are associated with broad categories. These categories are generally formulated and named in relation to regulatory standards (e.g. “good” is used for the lowest AQI category within the United States National Ambient Air Quality Standards). A “good” AQI category could be logically misinterpreted by the public to mean that the air quality in this range poses no health risk, rather than the more accurate description of a low health risk. Conversely, by the nature of their calculation, AQHI values (as well as category names) are intrinsically risk based and, thus, explicitly communicate to the public the lack of a safe lower threshold of air pollution exposure (i.e. “low risk” is the lowest AQHI category in Canada).

Despite the relative merits of AQHIs (Fig. 1), their implications for public health are not well described. A potential shortcoming is that all multipollutant AQHIs issued by governments to date (e.g. Canada) have been linked to one specific health outcome (e.g. mortality). Their ability to predict other health end-points (e.g. asthma or CVD events) may be less robust, given the differing concentration–response relationships for both a given pollutant and health end-point. Secondly, while the numerical values of AQHIs and AQIs may be discordant under similar environmental conditions (i.e. differ with respect to the percentile ranking within their own scale), the assigned categories will probably be mostly concordant (e.g. “low risk” for a Canadian AQHI versus “good” for a United States AQI) as they incorporate broad bands of values. This means that public messaging and behavioural advice may not differ between these indexes. However, it is conceivable that under some sets of conditions the combined effects of multiple pollutants (especially when levels are just below the regulatory thresholds) could elevate an AQHI but not an AQI into a higher category. It must also be remembered that the threshold values chosen for AQI and AQHI categories are subjective. Changes can and do occur, especially for AQIs when new national regulations are implemented. As such, any category discordance between indexes can change and, largely, reflects subjective threshold selections along continuous scales.

Fig. 1. Similarities and differences between AQIs and AQHIs



2.2 Existing and novel health-based AQIs

2.2.1 Canadian AQHI as an exemplar

In the early 2000s epidemiological evidence showed that adverse health impacts occur below the air quality standards used as thresholds for calculating the AQI categories in Canada *(50)*. This brought into question the validity of the approach to characterize air quality into simple categories (e.g. good, fair, poor) as a means to improve public communication and understanding. The epidemiological evidence, which increasingly focused on $PM_{2.5}$ and PM_{10}, could not identify threshold concentrations below which no harmful effects were observed in the population. In parallel, there was increasing recognition of the limitations of communicating air quality risk to the public via AQIs. A particular concern was that basing AQIs on the single most elevated pollutant surpassing its standard does not allow for the possibility that concurrent exposure to additional pollutants in the mixtures could further (e.g. additively) increase health risks *(46,51)*. It is also possible that the air pollution mixture may alter the health risks posed by individual pollutants (e.g. gaseous pollutants can increase the toxicity of $PM_{2.5}$) *(52)*. Public health communities and the Government of Canada also noted a significant shortcoming in the communication of air pollution risk to the public. Roughly 92% of premature morbidity (e.g. hospitalizations) and mortality associated with air pollution were occurring during periods when the AQI classified air quality as very good or good *(53)*.

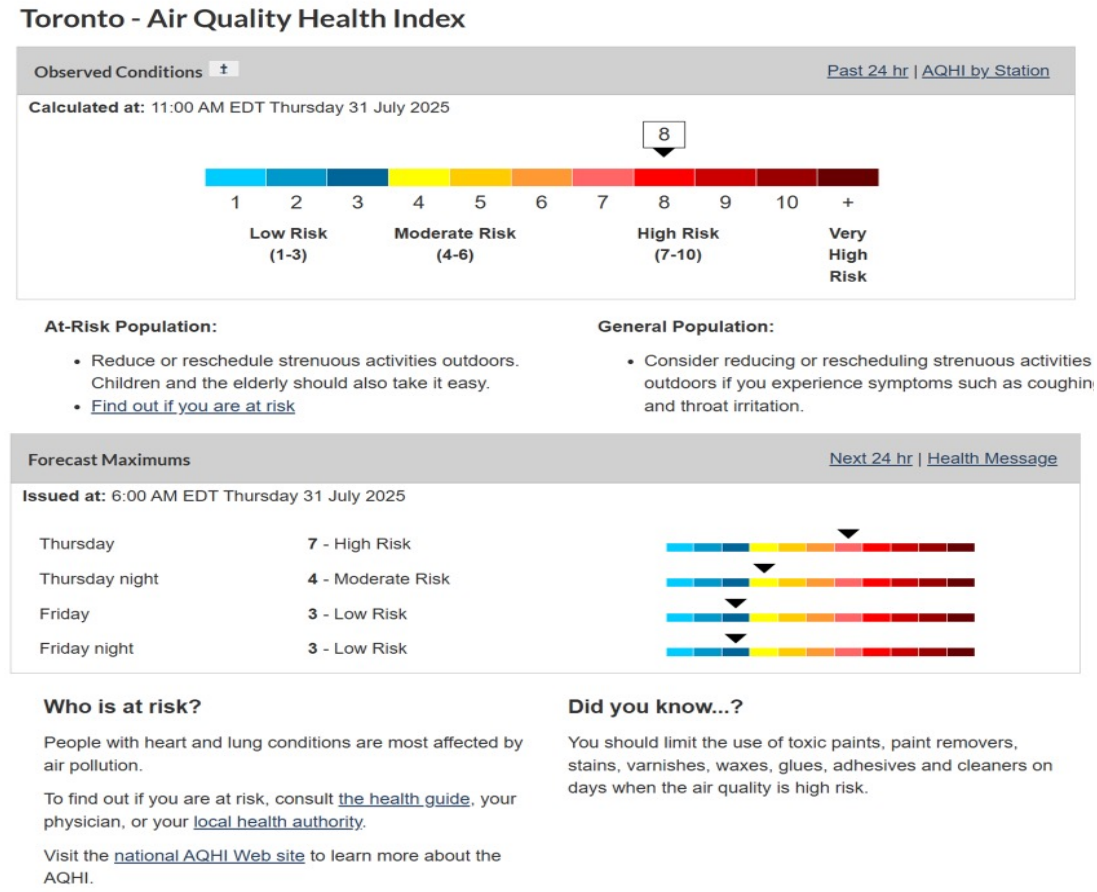
Therefore, a new AQI approach to address some key limitations was proposed by Health Canada (Canada's Federal Health Ministry) in collaboration with Environment Canada, representatives of provincial and municipal governments, nongovernmental organizations and other stakeholders. After a few years of local pilot programmes, in 2008 Canada became the first country to officially adopt a nationwide multipollutant AQHI using short-term air pollution levels and daily mortality data from cities across the country *(46,54)*. The modelling approach aligned with available epidemiological research showing that the combined effects of several pollutants on daily mortality consistently exceed the effects of the individual pollutants alone *(55)*.

The finalized AQHI was based on the concentrations of three pollutants: NO_2, O_3 and $PM_{2.5}$ *(46)*. The inclusion of CO and SO_2 was considered but rejected because their importance to health risk was not consistent across the country once the effects of the first three pollutants were taken into account. In practice, individual single pollutant models that quantify the excess mortality risks associated with ambient concentrations of NO_2, O_3 and $PM_{2.5}$ are combined additively into a single AQHI equation. The results are then transformed into a 10-point scale to facilitate public communication of the overall health risks and how they vary across the range of pollutant concentrations observed in Canada. A distinguishing feature of this novel approach is that the scale categories of 1–10 represent rising rates of predicted mortality risks (due to the combined effects of pollutants) and not simply increasing ambient pollution concentrations – with thresholds set by national standards. While all three pollutants play a role in the AQHI calculation, the underlying modelling is most driven by NO_2 concentrations in Canada. The merits of including several pollutants (i.e. the AQHI approach) versus the conventional AQI based on the pollutant that exceeds its standard by the greatest amount has also been demonstrated for selected United States locations *(56)*.

Stakeholder input was considered throughout implementation of the AQHI. Of note was Environment Canada's (Federal Environment Ministry) successful rollout of a national ultraviolet index and forecast system *(57)* and its priority of enhanced environmental prediction services. With comprehensive air quality models and greater computing power, the capacity for routine hourly, multiday, multipollutant predictions was becoming possible *(58)*. This meant that air quality, as represented by the AQHI, could be forecasted along with the traditional weather and ultraviolet radiation forecasts. These parallel objectives and the tendency for hourly air pollutant concentrations to be highly variable led to the AQHI being expressed as running 3-hour averages that could be updated throughout the day and predicted for future days.

Fig. 2 is a graphic designed to inform the public on how to interpret the Canadian AQHI and its forecast values. A challenge with communicating health risk is that there is a range of susceptibility across various populations. As a result, two sets of messages are provided: one for the general population and one for people at a greater underlying health risk (e.g. patients with asthma) *(59)*. People who believe they might be at higher personal risk are advised to obtain guidance from their doctor. This requires that the medical community is familiar with the AQHI, the adverse health effects of air pollution, and how to appropriately counsel patients at the various levels of susceptibility.

Fig. 2. Snapshot of the Canadian AQHI for Toronto at a given time



Air Quality Health Index Messages

This table is a summary of air quality health messages by category

Health Risk	Air Quality Health Index	Health Messages	
		At Risk Population*	General Population
Low Risk	1-3	**Enjoy** your usual outdoor activities.	**Ideal** air quality for outdoor activities.
Moderate Risk	4-6	**Consider reducing** or rescheduling strenuous activities outdoors if you are experiencing symptoms.	**No need to modify** your usual outdoor activities unless you experience symptoms such as coughing and throat irritation.
High Risk	7-10	**Reduce** or reschedule strenuous activities outdoors. Children and the elderly should also take it easy.	**Consider reducing** or rescheduling strenuous activities outdoors if you experience symptoms such as coughing and throat irritation.
Very High Risk	Above 10	**Avoid** strenuous activities outdoors. Children and the elderly should also avoid outdoor physical exertion.	**Reduce** or reschedule strenuous activities outdoors, especially if you experience symptoms such as coughing and throat irritation.

Note: the Canadian AQHI is a tool to be used by people to reduce their short-term exposure to air pollution and plan, on a daily basis, to modify their behaviour and reduce their personal health risk. (At-risk individuals/populations are determined through self-identification and/or informed by advocacy organizations or personal health practitioners.) The example shows the AQHI for Toronto at a given time. Step 1: identify the current (or forecasted) AQHI level for your location. The 1–10 scale (often stated as 1–10+ because air quality subsequent to the data period used to develop the formula and scale may have conditions (i.e. NO_2, O_3 and/or $PM_{2.5}$ levels) that lead to AQHI values of < 10) is categorized into qualitative descriptions of risk (low, medium, high and very high) according to different ranges of AQHI values: the higher the number, the higher the health risk. Step 2: identify your health risk status (i.e. susceptibility or sensitivity to air pollution). Information for the general population and susceptible groups can be obtained from multiple online sources. With the growth of the public's use of smartphones, applications that provide weather, ultraviolet radiation and air quality forecasts have been promoted for users to customize the information based on their personal air-quality risk level. Step 3: follow the relevant health messages *(59,60)*.

Source: Environment and Climate Change Canada *(61)* and Government of Canada *(62)*. Reproduced with permission from Environment and Climate Change Canada as a copy of the version available on the Air Quality Health Index and Air Quality Health Index Messages websites.

2.2.2 Additional strengths and limitations of the Canadian AQHI

The AQHI is inherently relevant for Canada because it was derived from current, locally relevant epidemiological data. While this is a strength, the underlying equation and risk categorization descriptions for the AQHI values cannot simply be adopted by other countries because the differing air pollution characteristics (e.g. pollutant levels, mixtures, sources, composition, toxicity) and population demographics (e.g. susceptibility, disease prevalence, secular event rates) can lead to different concentration–response relationships. Furthermore, how well the current AQHI (developed approximately 20 years ago) reflects variations in population risk in Canada is likely to change over time as air quality mixtures and demographics change (e.g. downward trends of some pollutants or increased prevalence of wildfire smoke). While updating AQIs as air quality standards evolve is relatively straightforward, updating the AQHI is more complex because of the need to repeat epidemiological analyses with newer data. Balancing such an update versus the need to communicate to users why the air quality appears to be different according to an updated AQHI is an ongoing consideration.

A potential shortcoming of the Canadian AQHI is that it was developed based solely on the daily mortality risk. Therefore, its relationship with specific health outcomes (such as pulmonary (e.g. asthma) versus CVD events) may be different and potentially less robust, given the differences in concentration–response relationships and in the importance of individual air pollutants. While many air pollutants often track together in a mixture, it is possible that on certain days, for example, only O_3 levels may be high. This could lead to a situation where the AQHI remains relatively low (because it is calculated based on excess mortality from three pollutants) but susceptible individuals (e.g. people with asthma) could, nonetheless, face unique health risks and are not made appropriately aware. Despite this limitation, the broader relevance of the AQHI has been somewhat validated in this regard. For example, the AQHI was found to relate to the usage of asthma health services *(63)*, emergency room visits for stroke *(64)* and asthma *(65)*, and 3-hour changes in ambulatory cardiac measures *(66)*. Its relevance outside large urban centres has also been evaluated *(67)*. A final limitation, which is not unique to the Canadian AQHI, is that the thresholds selected for rising categories (e.g. from low to moderate risk) are inherently subjective, given the monotonic concentration–response relationship. The simplified risk scale (1–10+) is intended to improve public communication. However, due to how the values are determined, there is no way to reverse translate into actual ambient pollution concentrations. No interpretations of the actual health risks, in either relative (e.g. 1% above ideal) or absolute terms, are conveyed. As such, there is potential for under-interpretation or over-interpretation of the health risks by individual members of the public. This represents an ongoing communication challenge.

2.2.3 Modifying the Canadian AQHI for wildfire smoke

Due to the impact of climate change and the increasing frequency of wildfires, there is growing concern about the responsiveness and applicability of the AQHI to wildfire smoke events, which are principally driven by high levels of $PM_{2.5}$ *(68)*. This led to the development of the AQHI+ formulation *(69)*, which is implemented when $PM_{2.5}$ concentrations are impacted by wildfire smoke. This tends to be when the $PM_{2.5}$ concentration exceeds 30 µg/m^3. In such wildfire events, the AQHI+ is calculated as the $PM_{2.5}$ concentration divided by 10 (irrespective of NO_2 and O_3 levels) and rounded up to the nearest whole number (i.e. a $PM_{2.5}$ > 30 µg/m^3 corresponds to an AQHI+ of 4 and is communicated as a moderate risk). Increasing the $PM_{2.5}$

concentration and AQHI from this level to high-risk values of > 60 µg/m³ and 7, respectively, represents another step because this hourly level correlates highly with a daily $PM_{2.5}$ concentration that exceeds the 24-hour objective for British Columbia of 25 µg/m³. For even higher $PM_{2.5}$ concentrations, the AQHI+ continues to increase monotonically with $PM_{2.5}$ concentration: the AQHI+ is 8, 9 and 10 when $PM_{2.5}$ equals 80, 90 and 100 µg/m³, respectively. Above 10, the AQHI+ is reported as 10+ regardless of the $PM_{2.5}$ concentration because guidance to the public does not change above this value (i.e. very high risk). This approach helps to address the unique challenges of degrading air quality from wildfire $PM_{2.5}$. It also reflects the fact that both morbidity and mortality risk correlate linearly with $PM_{2.5}$ levels, thus being consistent with the original risk-oriented approach of the AQHI. The AQHI+ is also applicable to the winter woodsmoke conditions related to frequent residential woodstove use for heating in British Columbia *(70)*. Given its relevance to such conditions and stronger relationship with respiratory outcomes, and also for consistency, the AQHI+ is now applied all year round in Canada. Both the AQHI and the AQHI+ are calculated and the higher of the two is communicated to the public simply as the AQHI. However, more research is needed to better understand the relative health risks of $PM_{2.5}$ from wildfires and other wood burning compared with fossil fuel combustion, given the differing compositions of their combustion products.

2.2.4 Novel AQHI approaches

The characteristics and potential benefits of employing a multipollutant AQHI have also been evaluated for other cities and countries worldwide (Table 1). A review of a range of multipollutant AQIs proposed that a consistent paradigm, the General Air Quality Health Index, would help to build a more accurate and comparable international AQI system *(35)*. A 2007 study proposed a multipollutant air pollution index derived from epidemiological data (i.e. directly linked to current health effect associations) *(71)*. The proposed index is linear and self-consistent, with a range of 0–10 set such that its value corresponds to the daily mortality risk associated with combined exposure to CO, NO_2, O_3, $PM_{2.5}$, PM_{10} and SO_2. This approach was evaluated for Cape Town (South Africa). A similar multipollutant index was proposed for Athens, Greece *(51)*; however, it is calculated based on constant thresholds derived from EU air quality standards, as opposed to epidemiologically based risk coefficients. Five pollutants (CO, NO_2, O_3, PM_{10} and SO_2) were considered but at least three are required, including NO_2 and PM_{10}, which are mandatory because of their importance to air quality in Athens.

Table 1. Overview of selected studies that have developed and assessed approaches for communication of short-term air pollution exposure risk that simultaneously considers multiple pollutants

Location	Outcome(s)	Pollutants	Reference
Chengdu, China	Cardiovascular mortality	NO_2, O_3, $PM_{2.5}$, SO_2	Xu et al. (2020) *(72)*
Guangzhou, China	Outpatient visits, hospital admissions, mortality	NO_2, O_3, $PM_{2.5}$, SO_2	Huang et al. (2022) *(73)*
China (272 cities)	Mortality	NO_2, O_3, $PM_{2.5}$	Du et al. (2020) *(47)*

Table 1 contd

Location	Outcome(s)	Pollutants	Reference
Ningbo, China	Mortality	Cardiovascular mortality: NO_2, O_3, SO_2 Respiratory and nonaccidental mortality: NO_2, O_3, $PM_{2.5}$, SO_2	Cao et al. (2021) *(74)*
Shanghai, China	Children's hospital outpatient visits for respiratory disease	Warm season: NO_2, O_3, $PM_{2.5}$, SO_2 Cold season: NO_2, $PM_{2.5}$, SO_2	Zhang et al. (2021) *(75)*
Tianjin, China	Cardiovascular, chronic respiratory and lung cancer mortality	NO_2, O_3, $PM_{2.5}$, SO_2	Wang et al. (2023) *(76)*
Tianjin, China	Cardiovascular-related years of life lost	NO_2, O_3, PM_{10}, SO_2	Zhang et al. (2024) *(77)*
Tianjin, China	Total mortality, years of life lost	CO, NO_2, O_3, $PM_{2.5}$, PM_{10} and SO_2 and combined in two principal components	Zeng et al. (2020) *(78)*
China, Hong Kong SAR	Respiratory and cardiac emergency hospital admissions	NO_2, O_3, PM_{10}, SO_2	Wong et al. (2013) *(48)*
India (2 cities)	Mortality	NO_2, O_3, $PM_{2.5}$	George et al. (2025) *(79)*
Tehran, Iran (Islamic Republic of)	Health (AQHI), safety and environment	CO, NO_2, O_3, $PM_{2.5}$, PM_{10}, SO_2	Ahmadi et al. (2015) *(80)*
Cape Town, South Africa	Mortality	CO, NO_2, O_3, $PM_{2.5}$, PM_{10}, SO_2	Cairncross et al. (2007) *(71)*
South Africa and global	Mortality	NO_2, O_3, $PM_{2.5}$/PM_{10}, SO_2	Adebayo-Ojo et al. (2023) *(81)*
Stockholm, Sweden	Asthma emergency room visits	Birch pollen, NO_x, O_3, PM_{10}	Olstrup et al. (2019) *(49)*
Bangkok, Thailand	Cardiovascular and respiratory outpatient department visits	O_3, $PM_{2.5}$, PM_{10}, SO_2	Kanchanasuta et al. (2024) *(82)*
Thailand	Mortality	CO, NO_2, O_3, $PM_{2.5}$/PM_{10}, SO_2	Punnasiri et al. (2025) *(83)*
Global	Children's respiratory hospital admissions or emergency department visits	NO_2, O_3, $PM_{2.5}$	Gladson et al. (2022) *(84)*
Global	Mortality	NO_2, O_3, $PM_{2.5}$, SO_2	Huang et al. (2025) *(85)*

NO_x: nitric oxide and nitrogen dioxide; SAR: Special Administrative Region.

Hong Kong Special Administrative Region (China) was the first additional location to follow Canada and officially adopt an AQHI *(48)*. This AQHI differs from the Canadian model in that it is calibrated based on emergency hospital admissions and includes four pollutants: NO_2, O_3, PM_{10} and SO_2. Many possible AQHI models have recently been developed as novel approaches, especially in the WHO Western Pacific Region (Table 1). However, official (or public) implementation appears to remain at local government level, such as in 2018 by the Government of Mexico City (Mexico) *(86)* and in 2024 by the Guangzhou Centre for Disease Control (China) *(13)*. Common issues evaluated in this research include the benefits of different formulations for different population groups (e.g. children, older people) and different seasons. To improve upon the additive approach of the Canadian AQHI for assessing the cumulative adverse health effects, different methods to combine the risks associated with multiple air pollutants have also been compared. The resulting AQHIs vary somewhat in their temporal behaviour and magnitude, as well as in their perceived responsiveness to health risk. The multipollutant indexes tend to correlate temporally but exhibit considerably different temporal characteristics than the conventional single pollutant AQI. Therefore, the former are considered to be better health risk indicators and communication tools.

Comparison of the range of outcomes and pollutants considered for AQHIs across cities, territories and countries shows that they are better than the single (maximum)-pollutant AQI because they are sensitive to health risks (Table 1). However, no clear single best approach (e.g. incorporated pollutants, modelling) that is useful in all locations has been identified. Owing to the importance of population susceptibility and relevant health outcomes, as well as air pollution concentrations and mixtures in their formulation, AQHIs best serve as public communication tools in those locations where they were developed. Therefore, the details of the AQHI calculation derived for a given geographical area and population should ideally not be applied globally for geographical comparisons of air quality and related health risk categories based on AQHI values. While a generalized approach has been proposed *(35)*, it is still expected to be calibrated on local data and, thus, AQHI values between different geographical areas would not be directly comparable. In demonstrating differences between two cities in India, another study argued that even more refined geographical AQHIs (i.e. city specific) may be needed to best protect the public *(79)*. Only two of the AQHIs listed in Table 1 propose a globally consistent formulation *(35)*. Since an approach developed and tested for South Africa set the thresholds and bands relative to current WHO guidelines, as opposed to local epidemiological data, it could potentially be adopted worldwide *(81)*. An AQHI calibrated to children's respiratory hospital admissions or emergency department visits also has potential as a globally consistent AQHI that includes NO_2, O_3 and $PM_{2.5}$, the same pollutants considered by the Canadian AQHI *(84)*. However, both approaches and other future options will introduce challenges, and these must be overcome before either can be considered a globally consistent approach to an AQHI. Such challenges include the wide range of local conditions among countries and/or cities (e.g. exposure levels, pollution mixtures, air quality management priorities) and potential variations in the underlying susceptibility of the population (e.g. variable demographics, population susceptibility, disease state frequency). The advantages of a health-risk-based (mortality) AQHI were recently explored across 12 countries/territories *(85)*. Using a consistent methodology in all locations and considering NO_2, O_3, $PM_{2.5}$ and SO_2, the study found that an advanced multipollutant model represented air pollutant mortality risk better than traditional AQIs or AQHI approaches based on single pollutant models. It demonstrated that the inclusion of NO_2 and O_3 is critical, more so than $PM_{2.5}$, thus highlighting the need for refined risk metrics. Lastly, a few novel indexes have been developed that further include the adverse health risks associated with additional environmental factors beyond air pollution such as extreme heat or elevated pollen levels *(49,87)*.

2.2.5 Novel approaches for behavioural advice

The current review failed to identify foundational evidence to justify the specific activity advice associated with current AQ(H)Is. Although some studies justify the actions, the behaviour changes largely represent common-sense, low-risk, easy-to-follow actions based on logical considerations. However, it is unknown whether they produce consistent and significant exposure reductions when executed at population level *(16,88–90)*. Staying indoors, which typically lowers O_3 exposure more than $PM_{2.5}$ exposure, or reducing physical activity (i.e. lowering the respiratory rate) can, in theory, reduce exposures to many outdoor air pollutants to variable degrees in many, but not all, people. However, this depends on several factors (e.g. indoor air pollution penetrance and sources, season, ventilation, outdoor concentrations) *(91,92)*. It is also important to note that, in general, people already spent most of their time indoors (80–90%). In principle, any action that aims to change the behaviour of a vast number of people must be extremely low risk for the ensuing benefit (i.e. reduced exposures) to outweigh any potential adverse consequences from the activity change itself or from taking no action.

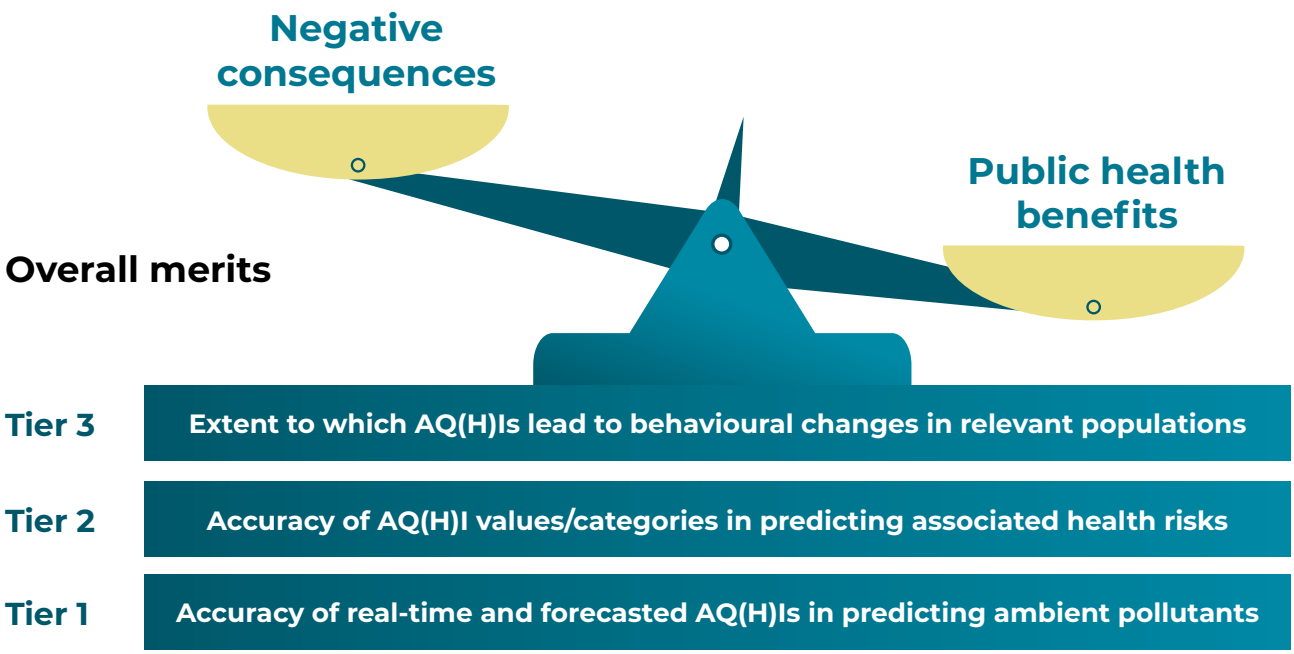
There is also no empirical evidence to drive the selection of specific breakpoints of air pollution levels (AQIs) or health risks (AQHIs) for issuing behavioural guidance for general population or higher risk groups (e.g. people with pulmonary disease or CVD). Given the monotonic concentration–response relationship without a lower safe level, the selected breakpoints have been largely tethered to air pollution levels set by national air quality regulations or expert opinion. They do not represent discrete air quality or health risk thresholds at which actions are abruptly warranted or have been proven effective. In this context, targeting more specific advice for population groups (e.g. based on better estimates of susceptibility) could, in theory, increase their effectiveness in reducing adverse health consequences. For example, interested users of the AQHI in Canada are suggested to self-calibrate by relating the current AQHI values to their own symptoms so that they know when to modify their behaviour. A recent modelling study demonstrated that the number of individuals in the general adult population, as well as patients with pulmonary disease, required to follow activity advice to prevent an adverse health event could be substantially lowered if the advice were better aligned to more targeted estimates of underlying disease-specific risk (i.e. atherosclerotic CVD, frequency of pulmonary disease exacerbation) *(93)*. Internet-based tools are widely available to estimate individual CVD and pulmonary disease risk. However, any potential benefits of such a precision or personalized approach would need to be weighed against the risk of making health advice overly complicated for the public.

2.3 Evaluation of AQIs and alerts

This report was commissioned with the goal of identifying roadmaps for best practices in developing, validating and communicating AQ(H)Is. To meet this objective, the expert writing group performed a comprehensive review of existing AQ(H)Is that focused on data from published studies that support specific strengths, weaknesses and limitations. The group first created a transparent approach to evaluate AQ(H)Is across three successive tiers that, together, influence the overall merit (Fig. 3). Tier 1 assesses their accuracy in reflecting air pollution concentrations in the location of interest. In Tier 2, the

associations between AQ(H)I strata with health risks are evaluated. Together, Tiers 1 and 2 represent what has previously been termed **validation**. Tier 3 considers the extent to which the issuance of AQ(H)Is (or air quality alerts) leads to appropriate behaviour changes in relevant populations. Lastly, the ensuing public health benefits (e.g. reduced mortality or pulmonary/CVD events) are estimated and can be (qualitatively) compared against potential negative consequences. This standardized evaluation framework can improve the scientific understanding of the merits of AQ(H)Is, identify specific limitations and help to elucidate potential pathways for improving existing AQIs. This report provides a narrative review of the evidence from key selected published studies relevant to each tier. The certainty of evidence was not evaluated (i.e. using the Grading of Recommendations Assessment, Development, and Evaluation criteria) because a formal systematic review was not undertaken *(94)*.

Fig. 3. Framework for evaluating the overall merits of AQ(H)Is and air quality alerts



The capacity of AQIs to accurately reflect the current or forecasted local ambient pollution exposures depends largely upon the accuracy, scale and sophistication of the monitoring network and exposure modelling employed to calculate the AQI *(15,95)*. Without adequate air pollution monitoring capabilities in place, it is impossible not only to implement effective air quality management but also to formulate accurate AQIs. Unlike in Canada, the United States and much of the WHO European Region (94% of its countries), over one third of countries worldwide have no large-scale or national air quality monitoring system *(95)*. The methodology of mathematically converting concentrations of air pollutants into AQI values is also a variable *(15,96,97)*. Since AQIs are typically formulated based on the (single) pollutant in a mixture that most exceeds its regulatory standard (e.g. $PM_{2.5}$), the value may not accurately reflect the concentrations of other pollutants that were less elevated on the day (e.g. NO_2, O_3). Lastly, the selection of threshold values for classifying air pollution levels into discrete AQ(H)I categories (encompassing bands of air pollution levels) is also important. This process inherently results in less-accurate exposure assignments for AQ(H)I categories than for individual AQI values. Nevertheless, in general, there is a moderate or strong

correlation between AQI values and air pollutant concentrations, which may vary by season, location and the pollutant driving the overall AQI designation *(98)*. Conversely, health-based AQIs (e.g. the Canadian AQHI) represent health risks (not air pollution concentrations) caused by exposures to multiple air pollutants. As such, AQHI values cannot be directly evaluated in Tier 1.

As AQI values indirectly represent the concentrations of the most-elevated air pollutant, it is logical to expect that they should significantly associate with health risks. However, their capacity to predict health risks depends not only on accurate air pollution measurements or exposure assignments (Tier 1) but also on the pollutant driving the AQI categorization and the health outcome assessed (e.g. all-cause mortality versus pulmonary/CVD events). Certain pollutants are more strongly associated with specific health risks (e.g. O_3 with asthma, $PM_{2.5}$ with CVD). In addition, the formulation of broad AQI categories inherently results in weaker correlation with health risks for AQI categories than for actual pollution levels (or individual AQI values), which have monotonic concentration–response relationships. This issue is most relevant in countries or regions with prevailing good air quality, that is, most days have AQIs in the lowest categories. For example, in the United States, there are significant short-term increases in health risks even when $PM_{2.5}$ concentrations are < 9.0 μg/m^3, despite all levels remaining within the "good" AQI category (i.e. "air pollution poses little or no risk") *(99)*. To a lesser extent, health-based AQIs (i.e. AQHIs) also have this shortcoming because their calculation captures the additive (or perhaps synergistic) health risks associated with multiple air pollutants, even at low levels. While $PM_{2.5}$ can have a dominant effect, there is mounting evidence that air pollution mixtures that commonly occur in real-world settings (especially, combinations of NO_2, O_3 and $PM_{2.5}$) can lead to greater health risks compared with exposure to any single pollutant *(100–109)*. The risk-based nomenclature of AQHI categories (e.g. low risk) also ensures they accurately convey the lack of a lower safe threshold in the concentration–response relationship. In summary, there is evidence from studies that both AQIs and AQHIs are associated with health risks in several different countries *(69,98,110–114)*. In a few studies that directly compared AQHIs with conventional AQIs, AQHIs performed better – probably because AQHIs capture the health risks from multiple air pollutants. Limited evidence further suggests that disease-specific AQHIs perform especially well *(76,111)*.

The extent to which the issuance of AQ(H)Is or air quality alerts is monitored and results in behaviour changes in the targeted populations has also been evaluated *(115–120)*. A key limitation of AQI thresholds for action, which vary substantially across countries owing to differences in the prevailing air quality, is that they are not empirically derived from trial or observational evidence. Typically, the activity guidance is tailored to susceptibility and designed to align with the severity of AQI elevation; it ranges from reducing outdoor activities/exercise or limiting time outdoors to staying indoors entirely. For some susceptible individuals (e.g. those with pulmonary diseases), the use of personal-protection devices (e.g. air cleaners, face coverings) during severe air pollution events (e.g. wildfires) has been suggested. However, it is important to note that there is a paucity of data showing the real-world effectiveness of these behaviour changes, especially at population level *(16,88–90)*. The effectiveness of staying indoors to reduce exposure varies among pollutants (e.g. O_3 versus $PM_{2.5}$) and depends on the season, outdoor concentrations, building aspects (e.g. indoor air penetrance, air-conditioning/filters) and indoor air pollution sources. In some locations, such as low- and middle-income countries, many households or indoor spaces may offer little protection from outdoor air pollution. Changing activities to minimize time spent outdoors, outside exercise (e.g. location, intensity, timing) or daily traffic exposure (e.g. commutes) will also have variable effectiveness. It must also be acknowledged that many people worldwide may have less agency to strictly follow activity advice (e.g. outdoor workers).

A systematic review of 21 studies concluded that there was moderate but suboptimal adherence to behavioural guidance issued by AQ(H)Is in the populations evaluated. A median of 31% of people reported avoiding outdoor activities and 46% reported taking other specific actions during poor air quality days *(118)*. Better adherence was associated with knowing how to check AQIs, believing that adverse symptoms are related to air pollution, the perceived severity of poor air quality and having received advice from health-care professionals. Recent national survey data in the United States also showed low overall concern about air quality and poor engagement with AQIs, with 60% of adults rarely (i.e. less than once per month) or never monitoring the AQI *(121)*. Older people and individuals with respiratory conditions are more likely to be concerned and alter their activity in accordance with AQI advice *(119,121)*. However, the findings were limited by the fact that most studies have been conducted in Canada, the United States and the United Kingdom, with none in low- or middle-income countries or regions (e.g. India, WHO African Region). In summary, the available evidence shows that less than half of the population routinely monitors AQIs and alters their behaviour in response to activity guidance or air quality alerts *(115–120)*. Individuals with chronic diseases (e.g. asthma) and older adults are likely to be more adherent to the guidance.

Perhaps the most crucial factor to consider when evaluating AQ(H)Is is whether their issuance (and the corresponding activity guidance) translates into a reduction in morbidity and mortality in the population. However, there are several limitations to consider. The precise threshold values for the AQ(H)Is selected for issuing behavioural advice are not evidence informed: they are not empirically derived from studies showing improved health outcomes by the public taking action to lower exposures when air pollutant levels (for AQIs) or health risks (for AQHIs) exceed the threshold values selected for providing activity guidance. For AQIs, the thresholds are typically based only on national regulatory standards (which vary across locations) for single pollutants. It is also not ethically possible nor viable to perform idealized investigations, such as randomized clinical trials with a control arm (e.g. no AQI information provided), to test the effectiveness of existing activity guidance to protect public health at current or differing AQ(H)I thresholds.

A literature review identified 12 observational studies that evaluated the health benefits of issuing AQ(H)Is (three studies) or air quality alerts (nine studies) based on varied methodologies (outlined in Table 2) *(115,122–132)*. Since the issuance of air quality alerts was generally based on air pollutants reaching high AQI thresholds specified at local level, they were included in the assessment. Five studies were conducted in the WHO Region of the Americas, with the results showing modest (or no) benefits from AQ(H)I guidance or air quality alerts. A 2022 modelling study using data from the United States calculated that tens of thousands to several million people (depending on baseline health risks and the outcomes assessed) typically need to follow the activity guidance to prevent a single ischaemic CVD (e.g. myocardial infarction, stroke) or pulmonary (e.g. asthma exacerbation) event in the population when air quality is in the "unhealthy for sensitive individuals" or "unhealthy" range for $PM_{2.5}$ *(125)*. Given the current air quality conditions in the United States, few adverse health events could be theoretically avoided per year by following AQI guidance *(133)*. The fact that levels of pollutants and thresholds for action are lower in the United States than in other global regions may partially explain these findings. Three studies were conducted in the WHO European Region. Two in the United Kingdom showed no benefit, or possibly even a potentially inappropriate increase in health-care utilization. A study in Paris (France) showed that instituting an alert programme at more stringent thresholds is associated with reduced CVD mortality. Four studies conducted in the WHO Western Pacific Region reported more encouraging results. Even so, only a subset of the multiple health outcomes evaluated were positively influenced by air quality alerts in Hong Kong SAR (China) and the Republic of Korea.

Table 2. Selected studies evaluating the health effects of AQIs and alert programmes

Location	Publication year	Study years	Pollutants and levels	AQI guidance or alerts and study method	Population/health outcomes	Main findings/conclusions
WHO Region of the Americas						
Longueuil, Quebec, Canada *(115)*	2019	2015–2016	Canadian AQHI: NO_2, O_3, $PM_{2.5}$ (levels not provided)	APWS issued warnings and advice when air pollution was in the moderate or higher scale of the Canadian AHQI	People vulnerable to air pollution (*n* = 662 treatment group; *n* = 666 control group)	APWS increased the awareness of pollution episodes and improved the adoption of behaviours to avoid exposures. There were no improvements in health outcomes, including smog-related symptoms and use of health-care services
				Odds ratios for outcomes comparing treatment to control group	Phone surveys for health questions	There was no evidence for health benefits from an APWS aligned with Canadian AQHIs
Toronto, Canada *(122)*	2018	2003–2012	Daily mean levels during study days: • NO_2: 20.7 ppb • O_3: 39.2 ppb • $PM_{2.5}$: 21.3 µg/m^3	AQI alert announcements: daily maximum AQI of ≥ 50 (1-hour O_3 of 81 ppb or 3-hour $PM_{2.5}$ of 46 µg/m^3)	Population cohort of ~2.6 million people	Alerts significantly reduced asthma-related emergency department visits by 25% (4.73 cases per million people per day). No other health outcomes significantly reduced
				Regression discontinuity design, effect of issuing AQI alert on outcomes	7 cardiopulmonary health outcomes	Issuing air quality alerts alone has a limited effect on public health. Measures focusing only on days with the highest levels of pollution are unlikely to help to reduce the burden of adverse health effects from air pollution in Canada

Table 2 contd

Location	Publication year	Study years	Pollutants and levels	AQI guidance or alerts and study method	Population/health outcomes	Main findings/conclusions
Denver, Los Angeles, Pittsburgh, United States *(124)*	2021	2014–2017	Daily O_3 and $PM_{2.5}$ (levels not provided)	United States AQI 6-level scale, with corresponding activity guidance	Adults aged ≥ 65 years	Less than 1 death per year was averted in all cities (using estimated mean outdoor pollution levels), even in models assuming zero indoor exposure by staying indoors
				Modelling study: mortality reduction by following AQI activity guidance for AQIs of 100 and 150 for O_3 and $PM_{2.5}$	All-cause mortality	Few deaths per year in older people can be avoided by following AQI activity guidance in the United States
United States *(123)*	2018	2007–2010	NHANES question on air quality	Response to NHANES question on any activity change if they thought or were informed that air quality was bad during the past year	NHANES: people with asthma versus those without asthma	People with asthma were roughly twice as likely to respond that they changed their activity if they thought or were informed that air quality was bad. Responses did not differ between people with an asthma attack or emergency visit for asthma in the previous year and the general population
				Survey logistic regression models to test whether people with asthma are more likely to change activities than those without asthma	Binary response to question about changing activities	It is plausible that awareness of poor air quality may lead people with asthma to make more activity changes that could prevent adverse health outcomes

Table 2 contd

Location	Publication year	Study years	Pollutants and levels	AQI guidance or alerts and study method	Population/health outcomes	Main findings/conclusions
United States *(125)*	2024	2022	Daily $PM_{2.5}$ levels: annual mean of 7.5 µg/m³ for the United States	United States AQI 6-level scale with corresponding activity guidance	Healthy adults and patients with ASCVD and pulmonary disease in the United States population	NNT was very high during the 96% of days when an AQI activity guidance was issued (AQI: 101–200) for the general adult population (> 18.4 million) and adults with ASCVD (~1.6–5.0 million) or pulmonary disease (~66 000–202 000)
				Modelling cross-sectional study calculating the NNT to avoid cardiopulmonary events by following AQI activity guidance for $PM_{2.5}$	Daily ASCVD and pulmonary events	Given the large NNT values, few cardiopulmonary events can be avoided per year by following AQI activity guidance in the United States; more pulmonary than CVD events can be avoided
WHO European Region						
Paris, France *(127)*	2021	2000–2015	Daily PM_{10} (levels not provided)	PM_{10} AQI alert programmes begun in 2008 (80 µg/m³) and revised in 2011 (50 µg/m³) for 24-hour periods	Entire population and adults > 75 years old	No reduction in mortality after implementing the 2008 alert programme. CVD mortality reduced 16% (386 prevented cases) in the population and 19% in people >75 years after implementing the 2011 alert programme. There was no reduction in pulmonary mortality following either alert programme
				Difference-in-differences approach with propensity-score matching to evaluate the effect of an AQI alert issuance on mortality	Daily mortality rates: total number of nonaccidental deaths = 622 268	Lower (but not higher) thresholds for AQI alerts reduced CVD mortality. It is important to validate the pollution thresholds for alerts based on empirical evidence for a benefit

Table 2 contd

Location	Publication year	Study years	Pollutants and levels	AQI guidance or alerts and study method	Population/health outcomes	Main findings/conclusions
Port Talbot, United Kingdom *(126)*	2016	2012–2015	Hourly PM_{10} (levels not provided)	Alerts by text, email or voice advised self-care and behaviour in 3 scales for 1-hour PM_{10} of ≥ 67 μg/m^3	Intervention group (*n* = 180): people with asthma, COPD or CVD	Alerts increased emergency attendances (IRR: 1.89), emergency admissions (IRR: 2.04) and respiratory emergency admissions (IRR: 3.97)
				Quasi-experimental study of alerts comparing relative changes in health-care utilization in intervention group vs those who did not sign up	Hospital admissions, emergency visits, health care contacts	Alerts were associated with increased (not reduced) usage of health services. Findings raise concern about trade-offs between increased harm vs benefits of air pollution alerting services
Southampton, United Kingdom *(128)*	2015	2008–2013	NO_2, O_3, $PM_{2.5}$, PM_{10} (levels not provided)	Alerts to subscribers via text, phone call, email or internet when local AQI was moderate or above for PM_{10} and O_3	197 subscribers to air quality alert system	Number of subscribers to the programme needs to be substantially increased to 5910 in the respiratory disease population to show a statistically significant reduction of 1 admission per year
				Likelihood of an admission caused by air pollution and comparing days where a pollution event occurred to when an alert was issued	Hospital admissions	Existing air quality forecasting alert service was ineffective at reducing hospital admissions. The cost–effectiveness of the programme was questioned

Table 2 contd

Location	Publication year	Study years	Pollutants and levels	AQI guidance or alerts and study method	Population/health outcomes	Main findings/conclusions
WHO Western Pacific Region						
China, Hong Kong SAR *(130)*	2019	2010–2016	Daily mean levels for study period: • NO_2: 49.2 µg/m^3 • O_3: 43.0 µg/m^3 • PM_{10}: 41.4 µg/m^3 • SO_2: 11.4 µg/m^3	AQHI alert programme began 30 December 2013. 5-point scale and guidance based on 3-hour mean of 4 pollutants	976 252 admissions for respiratory diseases (age standardized) during the study period	After implementing the AQHI alert programme, admissions for RTIs fell immediately by 14%. Decline in RTI was limited to people younger than 18 years; no decline in total or any other specific respiratory diseases
				Interrupted time series with Poisson segmented regression; effect of implementing the AQHI alert	Hospital admissions for respiratory diseases	AQHI alert programme may reduce hospital admissions for RTIs in children, but not any other respiratory disease for any age group
China, Hong Kong SAR *(129)*	2020	2010–2016	Daily mean levels for study period: • NO_2: 49.2 µg/m^3 • O_3: 43.0 µg/m^3 • PM_{10}: 41.4 µg/m^3 • SO_2: 11.4 µg/m^3	AQHI alert programme began 30 December 2013. 5-point scale and guidance based on 3-hour mean of 4 pollutants	375 672 hospital admissions for CVD in older people during the study period	After implementing the AQHI alert programme, admissions in older people for hypertensive disease and acute MI fell by 16% and 15%, respectively. No decline in total or any other specific CVD
				Interrupted time series with Poisson segmented regression; effect of implementing the AQHI alert	Hospital admissions for CVD	AQHI alerts may reduce admissions for some CVD events (hypertensive disease and MI) but not for total CVD or any other specific CVD in older people

Table 2 contd

Location	Publication year	Study years	Pollutants and levels	AQI guidance or alerts and study method	Population/health outcomes	Main findings/conclusions
Republic of Korea *(131)*	2021	2015–2019	Daily mean $PM_{2.5}$ levels for study period: 24.2 μg/m³	Mobile text messaging of WEA and AIT: 2-hour $PM_{2.5}$ levels: • watch: 75 μg/m³ (AIT only) • warning: 150 μg/m³ (both systems) Emergency reduction: 50 μg/m³ (2 days in a row) or 75 μg/m³ (next day)	318 591 individuals	Days with issuance of both alerts have 8.6% reduction in hospital visits and admission rates for respiratory diseases; the WEA programme was more effective at preventing outcomes (12.5%) than the AIT programme (2.5%)
				Generalized additive model of $PM_{2.5}$ related to health outcome, evaluating the effect of alert issuance	Daily number of visits and admissions due to respiratory sicknesses	Alert messaging using WEA (sent to all people with mobile phones) may be more effective than using an AIT (subscription only) for preventing respiratory health events
Republic of Korea *(132)*	2023	2010–2019	Daily median levels for study period: • CO: 532.9 μg/m³ • NO_2: 36.3 μg/m³ • O_3: 53.1 μg/m³ • PM_{10}: 44.0 μg/m³ • SO_2: 11.4 μg/m³	AQWS implemented January 2015	Outpatient or hospitalization claims from ~1 million people	After implementing the AQWS programme, there were no differences in exacerbation rates of any environment-related disease: COPD, asthma, stroke and CVD; an immediate decrease in incidence of CVD (34%) and stroke (43%); a gradual decline in asthma incidence (20%); and no change in COPD incidence
				Poisson segmented regression used to compare rates before and after implementing the AQWS	Age-standardized rates of incidence (diagnosis) and exacerbation (hospitalization or emergency visits)	Implementing the AQWS may decrease the incidence (diagnosis) but not the acute exacerbation of several diseases; there may be variations in response by disease and individual characteristics

AIT: air quality information text; APWS: automated phone smog warning system; AQWS: air quality warning system; ASCVD: atherosclerotic CVD; COPD: chronic obstructive pulmonary disease; IRR: incidence rate ratio; MI: myocardial infarction; NHANES: National Health and Nutrition Examination Survey; NNT: number needed to treat; RTI: respiratory tract infection; SAR: Special Administrative Region; WEA: wireless emergency alert.

Note: AQI guidance and study methods are shown in yellow; alerts and study methods are shown in grey.

Several reasons may underly the variable results of these studies. Differing experimental designs, health outcomes and statistical power to detect health benefits are likely to play a role. Passive advice associated with the daily issuance of an AQ(H)I appeared less effective than the active issuance of an air quality warning. However, there is some potential for confounding in this regard: alerts generally occurred during periods of very poor air quality, which itself may be responsible for driving the health benefits of public action. The comparative benefits observed in the WHO Western Pacific Region may be due to the higher prevailing air pollution levels (i.e. potential for greater health benefits of lowering exposures), use of active alerts in the relevant studies and/or greater public awareness of the associated health risks and importance of following alert advice. In summary, the available evidence shows limited public health benefits from issuing AQ(H)Is or air quality alerts in WHO Region of the Americas and WHO European Region. Conversely, there is evidence of health benefits related to issuing active air quality alerts for subsets of the public in two locations in the WHO Western Pacific Region.

2.3.1 Potential negative consequences of AQ(H)Is or air quality alerts

Air pollution is associated with many negative socioeconomic consequences. In some locations, air quality alerts are linked to regulations to limit or stop certain industrial activities over the short term. Such measures can be successful in reducing air pollution levels but may also result in negative economic impacts *(134–138)*. Direct adverse socioeconomic consequences from air pollution include loss of workdays and reduced productivity, school absenteeism and worse student performance, decreased outdoor and leisure activity, lower psychosocial well-being, and adverse impacts on tourism *(137–142)*. However, issuing AQIs and alerts is also hypothesized to alter public behaviour, resulting in unintentional harms such as a reduction in healthy behaviours (e.g. exercise), missed work or school, and greater exposure to indoor air pollution *(137–142)*. This may be especially relevant if low-risk individuals who are not specifically targeted by AQ(H)Is nonetheless alter their behaviour or if activity changes are not temporally aligned with poor outdoor air quality. Given that higher AQIs (and air quality alerts) are issued when the underlying air quality is worse, it is challenging to dissociate any potential independent negative effects of AQIs on socioeconomic outcomes from the direct harms of air pollution. The expert writing group could only find one relevant study conducted in Shanghai (China) *(142)*. It showed that the ostensibly harmless public advice to spend more time indoors when AQIs are elevated led to increased household energy consumption and citywide carbon emissions, which has the potential to offset some of the health benefits.

A theoretical concern is that by focusing solely on short-term air pollutant concentrations reaching relatively high levels, AQ(H)Is may contribute to a false sense of security in the public. For example, guidance for activity changes for the general public is issued in the United States only when the AQI reaches "unhealthy" levels ($PM_{2.5}$ of > 55.5 μg/m^3). Such high values are uncommon and isolated to a few locations: they occurred only 342 times across 3143 United States counties in 2022. This means that no activity change guidance was issued for 99.97% of days across all United States counties throughout 2022 (342 issuances/(3143 counties × 365 days) × 100) *(133)*. No guidance was issued for sensitive individuals for 99.9% of days ($PM_{2.5}$ of > 35.4 μg/m^3). It is plausible (albeit not yet studied) that the rare issuance of AQI activity change guidance could inadvertently serve to mislead the public into believing that air pollution poses no health risks most of the time. However, evidence shows that $PM_{2.5}$ is associated with significant risks when levels are well below these regulatory-based high threshold concentrations.

AQIs do not currently advocate for activity changes under most air-quality conditions in the United States. As such, they are not designed to protect the public against most of the adverse health effects of air pollution arising from repeated or chronic low-level exposures. This limitation is probably also true for many other countries with overall good air quality. To address this shortcoming, future AQIs may need to consider developing strategies to additionally communicate to the public the health risks related to long-term air pollution exposures.

Due to a paucity of evidence, no conclusions can be drawn regarding the relevance or importance of negative socioeconomic consequences arising directly from the issuance of AQ(H)Is. Overall, the review and assessment conclude that there is greater potential for public health benefits resulting from the issuing of AQ(H)Is (despite the limited available evidence), and especially for the active issuance of air quality alerts at higher air pollution levels, than for socioeconomic harm (Fig. 3).

2.4 Public communication of AQIs

2.4.1 General principles

Effective public communication strategies represent the critical bridge between public health risks posed by air pollution and generating individual-level motivation to take action, with the intent of reducing exposures. For interventions to successfully promote behaviour change, a one-size-fits-all approach should be avoided *(29)*. Communication strategies need to be tailored to fit aspects of individual communities (e.g. education, language) and respect local cultural norms. Messages should be specific enough for both the public and subgroups of susceptible individuals. For the health risks of air pollution to be seen as relevant by the public, there must be broad awareness, plus a general perception that the issue is of real-world importance for individuals in the community *(143)*.

Effective public communication that leads to health-protective behaviour begins with individuals accessing credible air quality information from trusted sources *(143,144)*. Once the public has been informed, behavioural adaptations may include limiting or avoiding strenuous outdoor activity, reducing time spent outdoors, avoiding high-pollution hot spots such as traffic-dense areas or peak commuting periods, and, during severe pollution episodes, staying indoors entirely. The use of personal protective measures – such as N95-equivalent respirators and portable indoor air cleaners – can further reduce exposure to $PM_{2.5}$ *(16)*.

Behaviour change that mitigates risk is a multistage process that begins with the intention or motivation to act *(145)*. Motivation arises from an individual's appraisal of the health risk (i.e. the perceived threat to themself), belief in the efficacy of the response (whether the intended behaviour will actually reduce risk), assessment of potential costs or inconveniences, and degree of self-efficacy (confidence in one's own ability to carry out the behaviour) *(143)*. According to socio-ecological models, three factors need to be present for behaviour change: capability, opportunity and motivation *(146)*. When any of these factors are lacking, a motivation–behaviour gap may emerge, whereby even motivated individuals fail

to act. Thus, even the most precisely crafted and well-timed messages will not succeed if individuals lack the physical or social opportunity to implement them. Effective communication must, therefore, occur within an enabling environment, that is, one that simultaneously addresses the social, physical and environmental barriers to action.

The perception of air pollution as a health risk may be one of the most important psychological factors to help an individual to successfully take action or execute a behaviour change *(147)*. Air pollution exposure may be construed as an amorphous threat that cannot be influenced at a personal level. However, communicating threats or harm through personalized air quality data can help people to make the connection between a change in behaviour and its impact on exposure *(29,143)*. Studies have demonstrated improved self-efficacy through protective action in response to monitoring data *(148)*. Lastly, it is important to acknowledge that behaviour change is also shaped by multilevel (not only individual level) influences that include societal norms and situational contexts *(143,149)*.

Risk communication approaches need to be flexible enough to accommodate and communicate alterations in risk that may result from differences in air pollution mixtures and baseline concentrations. Cultural differences may also drive the perception and tolerance of risk associated with environmental factors and relevant behaviour change *(29)*.

The use of technical and impersonal language is a frequently cited challenge that results in poor understanding and/or ineffective public communication of the personal health threats posed by air pollution *(150)*. This includes ambiguous messaging on air quality being "good", as designated by many AQIs. Another common misperception is that access to publicly communicated information is sufficient to promote behaviour change or motivation to alter behaviour (i.e. protection motivation). Unfortunately, the mere receipt of air pollution alerts may not translate into appropriate behaviour change.

2.4.2 Communication approaches

Effective communication about air pollution should be viewed through an ecological lens, in recognition that communication is not merely about changing individual behaviours but also about shaping the social, cultural and policy environments that enable collective action and systemic change. Communication strategies can influence how communities understand their shared environment, build trust in institutions and mobilize public support around clean air initiatives. Within this broader context, the dissemination of air quality information becomes a tool not only for personal risk reduction but also for fostering environmental stewardship. Standards for the effective communication of air pollution and other environmental risks have been articulated by numerous health agencies *(15,29)*. Nevertheless, there is heterogeneity in how air quality risks are reported and perceived by the public across the world *(29)*. Recent technological breakthroughs mean that real-time and forecasted air pollution levels are now widely available to the public *(151)*. Mass media sources (e.g. newspapers, government (public) and private websites) are recognized as a source of credible information about air quality. However, extensive evidence suggests that a lack of quality information can perpetuate public misperceptions about health risks *(152,153)*. Media framing theory explains that how the news is structured (i.e. wording, narratives and visuals) shapes public understanding of events *(154)*. In air quality reporting, an emphasis on certain details can skew perception. Dramatic imagery or alarmist language may heighten fear and pessimism, whereas

subdued coverage can minimize the perceived risk *(155)*. When framing is intentionally distorted – such as by exaggerating policy successes or downplaying air pollution data – this can create false optimism, confusion and emotional polarization. A recent study from China investigated how misleadingly optimistic reports of cleaner air quality affected both public perception and actual air quality *(156)*. Using a combination of global data analysis, behavioural experiments, field studies and agent-based modelling, the study found that less press freedom was associated with less accurate air pollution reporting and higher air pollution at global level. In experimental and field settings, exposure to misleadingly optimistic air quality reports reduced individuals' perceived air pollution levels and significantly increased their willingness to travel. Agent-based modelling further showed that such behavioural shifts collectively worsen air quality by increasing emissions as a consequence of changes in behaviour. Overall, the study suggests that misleadingly optimistic air quality reports can trigger behavioural feedback loops that degrade real-world environmental conditions, thus underscoring the critical need for accurate, transparent environmental communication in China *(156)*. It must be emphasized that, given the current global debate on the role of social media and perpetuation of misleading information, there is a need for trustworthy sources of information. Smartphones may enable flexible and personalized engagement; however, care must be taken to avoid shifting responsibility to the individual from national and local authorities *(157)*.

Existing air quality communication strategies lack critical information, including specific evidence-based risk mitigation behaviours (information quality), and often fail to reach vulnerable populations, mainly owing to technological and language constraints (information reach) *(158)*. Research on health promotion campaigns indicates that the most persuasive communication method is to complement negative descriptors of poor air quality by promoting pathways that enable people to protect themselves and mitigate the health risk associated with poor air quality *(159)*.

In summary, the implementation and effectiveness of air pollution communication vary widely. Misinformation and distorted framing could produce real environmental harm through behavioural feedback loops. Therefore, communication about air pollution must be accurate, transparent and contextually sensitive, avoiding both alarm and false reassurance. Future strategies should ensure equitable access to credible information, clearly outline actionable protective behaviours and foster public self-efficacy (belief in one's ability to take action and achieve a goal), while maintaining accountability among institutions responsible for environmental governance and risk communication.

2.4.3 Public awareness and risk assessment of air pollution

Public awareness of AQ(H)Is, and air quality in general, informs risk perception. In addition, accurate knowledge and risk perception regarding the health effects of air pollution play a central role in shaping individuals' responses and, ultimately, influences the capacity for AQ(H)Is to protect public health *(147,160)*. Although data show that people recognize the seriousness of environmental risks (including air pollution), they do not generally perceive them as a priority, especially in their daily lives *(121)*. Environmental risks are viewed as less dangerous and less urgent than risks to the economy or other more direct threats to health (e.g. accidents). Several sociodemographic characteristics (e.g. ethnicity, urbanicity) and health status can influence the public's health perceptions regarding air pollution *(149)*. Studies also show that prevailing air pollution concentrations, daily air quality changes and the frequency of AQI elevations also impact public awareness and perception of air pollution as a risk factor for health *(36,161)*.

Considerable variation exists in the scales used to measure the multiple dimensions of risk perception for air pollution. The domains most often investigated are "perception/awareness of air quality" and "perception of health risk". In a recent review of 95 studies, only nine studies (9.5%) used validated scaled questionnaires that assessed the perception/awareness of air quality and health risk *(162)*. There is also wide variation in the percentage of people who perceive air pollution as the most important environmental risk. Even if people recognize the seriousness of the risk related to air pollution, they may not necessarily perceive it as the most important issue in their country *(29)*. Nevertheless, several large natural experiments with longitudinal observations suggest that air quality alerts can lead to a reduction in outdoor activities *(115–120,157)*. The most-studied behaviour modifications have been reducing outdoor activity, time spent outdoors or specific activities such as driving; or protective or avoidance behaviours in busy streets, such as changing commuting times to avoid peak periods, closing windows, changing air filters or wearing face coverings. A review found that many large cross-sectional surveys of self-reported behaviour change or behaviour intentions support changes in behaviour, with a few exceptions *(157)*. Lastly, a recent narrative review of studies published from 1999 to 2024 on the public perception of air quality and AQIs in many countries showed that, while improvements are being made, there remains inadequate awareness of, trust in and personal action taken in response to official AQIs *(163)*. These problems are more challenging in low- and middle-income countries where air quality and the public health burden are worse.

2.4.4 Developing effective communication strategies

Key aspects of effective communication are the information sources, quality of the information and extent of reach, especially to vulnerable populations and susceptible individuals *(158)*. Several studies have highlighted the important role of trusted information sources, such as doctors and community health professionals *(144,164,165)*. Information is often better received by the public from these sources than from local or national government. Linking individualized, context-specific exposure insights with inclusive participatory frameworks can strengthen public engagement and promote sustained behavioural adaptation *(166)*. Interactive participatory research methods can translate into more effective behaviour change because such methods require contextual knowledge of the local population, which is best uncovered through collaboration and community engagement *(167–169)*. It is important to ensure that community members are able to engage in education or training in air quality activities, participatory modelling and reviewing policies with statutory organizations *(150)*. Instead of developing interventions for people, this approach develops interventions with relevant stakeholders so that they fit the needs and priorities of the people they impact *(150)*. Co-production of interventions designed around the local context and priorities is only now being realized in air pollution mitigation strategies.

Providing practical, contextual, culturally relevant and actionable information to individuals can help them to increase their self-efficacy regarding health risks and feeling empowered to act. Self-efficacy is a critical component in motivating behaviour change *(157)*. Communication of air pollution risk, together with a range of possible corrective behavioural actions, increases the chance that an individual can respond appropriately *(157)*. There is evidence that supportive behaviours such as advocacy and local community actions, which are not typically emphasized, can be highly effective in creating enabling environments for policy change *(170)*. There is also an opportunity to mix traditionally suggested health avoidance behaviours to reduce exposure with active co-benefit strategies such as walking and cycling

instead of driving, which increases physical activity. Except in extreme air pollution scenarios, such strategies may provide overall health benefits despite the increased air pollution exposure (but stops being net-beneficial for sensitive populations (aged > 60 years) at moderate AQHIs), while avoiding transportation costs, reducing congestion and cutting greenhouse gas emissions *(171)*.

Digital strategies and social media have significantly enhanced air pollution communication by broadening its reach and improving its relevance, especially for vulnerable groups. Social media platforms, text messaging and interactive digital applications deliver targeted, real-time alerts and empower individuals to take protective actions, while encouraging information-sharing within communities *(172)*. Artificial intelligence-powered approaches have the potential to further personalize risk information and forecast local conditions by optimizing messaging for self-efficacy and adaptive responses *(173,174)*. These innovations have the potential to promote inclusive, rapid and participatory communication so as to maximize behaviour change and public health protection.

2.5 Global equity

The global equity implications of AQ(H)Is and air quality alerts remain insufficiently addressed. This contrasts with the fact that AQ(H)Is are meant to specifically target susceptible and vulnerable populations, including people living in many low- and middle-income nations, socioeconomically disadvantaged groups, children, older people, and people with pre-existing cardiopulmonary diseases *(175,176)*. In higher-income nations such as the United States, vulnerable communities include ethnic and racial minority groups. Many vulnerable individuals reside in high-exposure microenvironments (near traffic corridors or industrial sites or in poorly ventilated housing) and/or may be homeless. This limits their capacity to receive and respond to standard air quality alert behavioural guidance, such as staying indoors or reducing physical activity. As climate change contributes to an increasing frequency of poor air quality episodes worldwide, AQ(H)Is and air quality alerts will increasingly become a critical adaptation mechanism to mitigate short-term health risks from exposure *(177)*. As such, equity issues are critically important to consider throughout the process of developing, validating and communicating health-based AQIs.

Current AQ(H)Is and air quality alert frameworks suffer from a one-size-fits-all risk communication strategy, including for vulnerable subpopulations. They do not adequately address disparities in digital access, (environmental) health literacy and linguistic diversity, or the occupational realities of outdoor workers who cannot feasibly reduce their exposure. Furthermore, none have systematically embedded equity frameworks. To advance equitable protection to vulnerable populations, AQ(H)Is and alerts must integrate some or all of the following adaptation policies:

- measures to communicate AQH(I) effectively to vulnerable populations, including those most at risk such as homeless people, those who lack access to clean indoor air (which is common in low-income countries) and those with limited ability to adapt to high air-pollution levels *(177,178)*;
- reducing private transportation in crowded urban areas, especially during high-exposure periods;

- economic subsidies such as for indoor air purifiers to vulnerable populations during high-exposure periods or access to communal centres or shelters with clean indoor air *(178)*;
- occupational health regulations to protect outdoor labourers during pollution episodes; and
- multilingual, culturally sensitive and accessible public health messaging.

Clean air is a universal human right. Recognition of this right underscores the need for globally uniform air quality standards, such as the *WHO global air quality guidelines (5)*. Uniform standards should also allow for regional adaptations based on local pollutant concentrations and socioeconomic considerations, while ensuring that all populations receive timely, equitable protection from the health threats posed by air pollution. Therefore, it could be argued that the development of uniform AQ(H)Is that are standardized and relevant for everyone living anywhere in the world should be considered *(15,179,180)*. Efforts to standardize AQ(H)Is and air quality alert frameworks hold significant potential for advancing global health equity and facilitating transnational comparisons of risk. However, as underscored by recent research, a universal one-size-fits-all approach remains inadequate for several reasons, including a limited ability to extrapolate AQHIs to populations and locations other than those where they were derived *(15)*. Furthermore, a uniform approach may not acknowledge the deeply rooted disparities in exposure, adaptive capacity and vulnerability that persist across and within countries *(177,181)*. True equity in the protection offered by AQ(H)Is will be realized only when these systems systematically account for differences in pollutant composition, local emission sources, societal infrastructure and the lived reality of marginalized populations, including those impacted by structural inequities, limited digital access, precarious housing or inflexible occupational demands. Only by integrating these considerations can global AQ(H)Is and alerts fulfil their foundational mandate to provide timely, actionable protection for all, especially those most at risk in a rapidly changing climate.

3. Key considerations and roadmaps for future AQ(H)Is

This chapter discusses issues to consider related to the development, validation, communication, further refinement and global equity of AQ(H)Is, which arose from the review of the published literature (Chapter 2). Concise discussions and insights are provided under each consideration based on the consensus of the expert writing group.

3.1 Development

> **Consideration 1:** define and rationalize the basis for AQ(H)Is – should the focus be on providing information to the public on local air quality (i.e. right to know), health risk communication, or both?

AQ(H)Is serve a critical role in informing the public and raising awareness about air quality and related health effects. However, only limited evidence that issuing activity/behavioural advice improves public health was identified. The paucity of research and difficulty in conducting studies are at least partially responsible for this. Nonetheless, providing practical advice to help to reduce short-term exposures is a relevant component of AQ(H)Is and should remain within their scope/mission. Withholding this guidance does not offer any apparent advantage and may only serve to diminish the knowledge and agency of the public to take safe, simple protective actions. In theory, it could also heighten public alarm regarding air pollution because exposure and health risks may be viewed as unavoidable.

It should be acknowledged that issuing AQ(H)I advice inherently relies on a supposition that factors assessed at population level (i.e. exposures, health risks) can be validly/accurately applied to small groups or even to single individuals and, thereafter, translated into effective personal-level advice. Such a population health strategy requires several extrapolations but is by no means unique to AQ(H)Is; as examples, it also applies to nutrition guidance based on epidemiological data and to medication indications based on clinical trial results. Similar to many other conditions (e.g. hypertension), the health risks from air pollution exposure and benefit (or lack thereof) from treatment (i.e. following AQ(H)I advice) are properties inherent to (and can only be determined for) groups of people. The summary risk estimates developed from cohorts of people are extrapolated to single individuals, but the actual risk applicable to any single person is not quantifiable (i.e. if individual risk, per se, can be said to exist at all) *(182)*.

Another concern is that making changes to an established AQ(H)I system requires significant resources, effort and time. Changes may also come with unintentional or unexpected drawbacks, such as confusing the public and/or fostering a misperception that the scientific evidence supporting adverse health effects of poor air quality is changing or even unreliable. Modifications viewed as likely to be positive may still face initial setbacks and/or take significant time to realize benefits. In this regard, careful considerations of the net benefits, costs and plausible pitfalls are warranted when considering modifications to AQ(H)Is.

Consideration 2: define the merits and weaknesses of multipollutant AQHIs versus conventional AQIs – should AQHIs be favoured over AQIs?

The review concludes that multipollutant AQHIs (i.e. health based) have significant strengths over conventional AQIs (i.e. regulations based). As such, governments should consider moving towards developing and employing local (i.e. country-specific) multipollutant AQHIs whenever possible. It is recognized that this may not be a viable option in many locations owing to resource or monitoring network limitations and/or lack of availability of the necessary health data. In this case, there are reasonable AQI modifications or interim approaches to consider (discussed further in Consideration 6). The lack of studies showing public health benefits of transitioning to an AQHI system is also acknowledged, despite the many putative advantages.

Consideration 3: define the strengths and weaknesses of issuing a single unitless AQ(H)I value.

Communicating a single easy-to-understand AQ(H)I number is a reasonable approach to provide clear and concise messaging regarding air quality to the public. However, it inadvertently results in a lack of transparency in potentially useful information. Unitless AQHI values cannot be interpreted as precise increases in health risks. The extent to which the public understands the relative or absolute risks posed by short-term air pollution exposures remains unclear. For example, one could reasonably misconstrue an AQHI value of 3 to mean that health risks are increased by threefold or by 3% compared with ideal conditions (whereas neither are true). Therefore, it is important that the health risk estimates for AQHI values should be made accessible to the public as part of the system. A straightforward example is: "today's AQHI value represents a 1.0% increase in mortality risk for the general adult population above the cleanest air quality conditions in your location". The most effective approach to communicating this complex issue along with the meaning/context of the health risks to the public requires input from many stakeholders. Nevertheless, this would represent a positive step forward by providing greater transparency of information and empowering individuals to make more-informed decisions.

It is also important to report the concentration of each measured air pollutant to the public in addition to the overall AQI or AQHI value. There is no clear rationale for denying the public ready access to the full set of measured air pollution information. Where an AQI system is still employed, then either the AQI level or concentration of each measured pollutant should be provided, in addition to the pollutant with the highest AQI. Following daily changes in multiple pollutants, or one specific pollutant, may be helpful for sensitive individuals (e.g. people with asthma, who are susceptible to high O_3 levels). For example, it is possible for an overall AQHI to not adequately reflect personal health risks if the multipollutant AQHI for mortality is low (due to low levels of NO_2 and $PM_{2.5}$), despite O_3 levels being relatively high and

placing sensitive individuals at risk of asthma exacerbations. Knowing the dominant pollutant may also be relevant (e.g. $PM_{2.5}$ during wildfires). As an example, the United Nations Environment Programme's global monitoring network provides $PM_{2.5}$ concentrations for each location *(183)* (discussed further in Consideration 6).

A tiered information model would be useful for the above-mentioned scenarios, whereby users could access increasingly detailed data, as relevant (e.g. in a footnote or by clicking/hovering on AQHI values in internet-based platforms). However, it is acknowledged that risk perception and interpretation are complex issues. Educational tools would need to be developed in partnership with community and other stakeholders and then incorporated into the system. Practical means to assist in better understanding the context of the risk increases could be provided – such as relating them to other common factors (e.g. acute stress, traffic accidents, poor diet, lack of exercise, tobacco smoking). However, developing the most effective methods requires more research. It is also unclear whether the availability of this added information would improve the public's awareness of air quality, understanding of the health risks and adherence to behaviour changes, or yield better health outcomes. Nevertheless, these efforts are aligned with the over-arching principle that the public has a right to full transparency and ready access to all measured air pollution information, as well as the related health risks *(184)*.

> **Consideration 4:** define and rationalize AQ(H)I scales and thresholds/bands of risk categories, given the monotonic concentration–response relationships for pollutants (no stepped risk or safe threshold at population level); and translating risk estimates derived from large-scale population studies to individuals or subgroups with varying susceptibility.

The process of segmenting continuous data (e.g. concentration–response relationships) into subjective bands or categories has inherent limitations. Nonetheless, this strategy is reasonable for AQ(H)Is because it enables clear and concise messaging for the public on air quality, health risks and behavioural advice. Although pollutants are monotonically associated with health risks, the real-world variations in adverse health effects across a single AQ(H)I category (e.g. from 1 to 3 in the low-risk category of the Canadian AQHI) are modest and, thus, reasonably warrant the same behavioural guidance. Moreover, a range of specific AQ(H)I values are also provided to the public within each broad category, enabling people to make more refined health decisions or further self-calibrate (e.g. per Canadian AQHI) to their own perceived risks. However, it is important to not lose focus on the over-arching goals of AQ(H)Is, that is, to inform the public regarding air quality and related health risks and provide practical options to help to reduce exposures. Their potential for success presupposes reasonably accurate exposure (AQIs) or health risk information (AQHIs). However, it is clear that there are diminishing returns in public health benefits that can feasibly arise from communicating increasingly precise exposure and health risk estimates. Minute differences in this regard (e.g. representing variations of a few micrograms per cubic metre in $PM_{2.5}$ concentrations) are likely to be of questionable health relevance, not easily understood by the public, and not translatable into meaningful changes in advice. Achieving an appropriate balance of communicating accurate but useful (i.e. understandable, practical, feasible) and not overly precise/complex information to the public is fundamental to the capacity of AQ(H)Is to meet their over-arching objectives.

The approach used by the Canadian AQHI in this regard is sound and can serve as a model. First, transforming data into a simple 10-point scale (as also employed by the daily AQI in the United Kingdom) presents information in a more relatable and intuitively understandable style compared with many AQI scales (e.g. with a range of 0–500). A downside may be that the narrower range of the scale does not provide an opportunity to convey as much detail about small changes in pollution exposures (or health risks) than a wider scale (e.g. 1–50 for a "good" AQI versus 1–3 for a low risk AQHI). However, minor changes in health risks, such as those occurring between discrete AQHI values from 2 to 3, are likely to be of little real-world importance in the context of informing and providing protective advice for the public. Secondly, the AQHI range (i.e. lowest to highest value) is inherently configured to be attuned (or standardized) to health risks from air pollution in the location where it was developed. As such, day-to-day changes across the AQHI scale are more discernible than those of most AQIs and provide more relevant/useful information to guide activities in relation to the air quality conditions actually faced by the local population. Conversely, more than half of the values across the AQI scale range (e.g. 200–500 in the United States) are of little contemporary relevance because they represent air pollution levels that rarely (if ever) occur in many locations worldwide. Thirdly, the bands/categories are configured across the AQHI scale such that they represent stepwise and uniform elevations in health risks (rather than arbitrary non-uniform elevations in pollutant levels, as for AQIs). This allows for a more cogent alignment of activity advice within each category to regular increments in health risks. Lastly, the method for aligning activity advice with individual susceptibility (i.e. people with pulmonary disease) with the additional option to self-calibrate risk is reasonable. Strategies to further refine this approach are discussed in Consideration 17.

> **Consideration 5:** define/refine activity guidance in AQ(H)Is, given the low absolute acute health risks and uncertain exposure reductions/health benefits of actions.

The incremental increase in absolute risk for adverse health effects (e.g. mortality, emergency department visits) due to air pollution exposure over a single day is small. Thousands to even millions of people must follow behavioural advice each day to prevent a single adverse health event. In this context, it follows that behavioural actions to reduce exposures must remain simple, low cost and low risk for the public health benefits to outweigh the adverse impacts of taking no action. The common-sense behaviour changes currently advised by many AQ(H)Is have the potential to reduce exposure, and accord with this general principle. Their continued issuance is reasonable, especially given the current lack of proven alternatives. Conversely, the cost–benefit and risk–benefit ratios of more intensive actions are less certain (e.g. evacuations, clean air shelters, air cleaners, face coverings). While small studies show potential health protection (e.g. from improvements in biomarkers), the broad public usage of such actions is not appropriate until more evidence is available. Nonetheless, it is reasonable for highly concerned susceptible people (e.g. patients with asthma or CVD) to choose to use one or more of these actions on a case-by-case basis (i.e. to self-calibrate) during severe pollution episodes such as wildfires. The lived experience, which varies substantially across locations (e.g. cities, regions, countries) and cultures, also impacts the awareness of and capacity to follow behavioural advice (e.g. education, income, access to digital information, household characteristics, work setting). An improved understanding of these factors and how to overcome barriers is important. More research is needed to enhance AQ(H)I behavioural guidance in the future, for example, to develop more evidence-informed thresholds for issuing advice, evaluate the comparative effectiveness and cost–effectiveness of different actions, and improve public communication, awareness and adherence.

> **Consideration 6:** improve methodological approaches (e.g. modelling, health-specific outcomes, combining multipollutant risks) during the development and validation of AQHIs – what is the evidence from global studies?

The review of studies on developing AQHIs shows that a range of methodological improvements on the Canadian AQHI could be considered. Various possible refinements of the formulation of AQHIs include involving disease-specific outcomes (e.g. pulmonary diseases, CVD) and targeting sensitive segments of the population (e.g. children, older adults) or more precise geographical locations. Refining the methodology to combine the risks from multiple pollutants is also possible, for example, the additive application of single pollutant models or use of multiple pollutant models that use different approaches to address collinearity. However, there is a risk that such changes would increase the complexity of the information communicated to the public and may outweigh the putative health protection benefits derived from adopting the new approach that may be scientifically superior but still imperfect (i.e. due to adopting many underlying assumptions). Therefore, simplicity/clarity for users and feasibility for widespread application must also be considerations in AQHI development. No single universal approach is likely to work for all locations. Individual countries need to adopt the most feasible and relevant approach for their location by balancing the costs, monitoring and modelling capabilities, and most important health outcomes.

In committing to the development and application of a multipollutant AQHI (outlined in Consideration 2), it is critical to be aware of resource implications. Crucially, these include increased requirements for air pollutant monitoring networks because not all relevant pollutants are regularly monitored and the effort to add them with acceptable levels of accuracy and precision varies across countries. The resource implications of being able to forecast future multipollutant AQHIs are also important to consider. Not all jurisdictions have the capacity to routinely run comprehensive air quality prediction models and interpret their output or to develop comparable empirical prediction models. In such contexts, an AQHI based solely on $PM_{2.5}$ levels may be a reasonable option (or interim step), especially in settings that lack the capacity to monitor multiple daily pollutants. Given the more widescale feasibility to monitor $PM_{2.5}$ levels, this approach could provide near-global coverage (e.g. by incorporating daily satellite-based concentration estimates). Since this approach has been evaluated in studies and AQHI models currently include varying numbers of pollutants, formulating a $PM_{2.5}$-AQHI is plausible. A $PM_{2.5}$-AQHI covering all geographical areas could be supplemented by more-refined multipollutant AQHIs whenever possible. If changing to an interim $PM_{2.5}$-AQHI is not a realistic option, then relatively simple but potentially useful modifications to conventional AQIs can be made. These include changing the lowest category name, where relevant, from “good air quality” to “low risk” (to avoid misleading the public) and providing the concentrations or individual AQI levels of all air pollutants measured (rather than the overall AQI).

Consideration 3 discussed the rationale to provide individual pollutant concentrations in addition to a single AQ(H)I value. An alternative (but more complex) strategy would be to develop models and provide AQHI subcategories that reflect the risks related to disease-specific health outcomes. This may particularly benefit sensitive populations, such as people with asthma or with a high risk for CVD, for whom individual health risks (i.e. asthma attacks or CVD events) may not be adequately captured or accurately reflected by the overall AQHI, which is calibrated for a different primary outcome (i.e. mortality). Although a few studies have shown the merit of this approach (reviewed in Chapter 2), it could be difficult to formulate in locations without adequate pollution monitoring or detailed health outcome data.

> **Consideration 7:** how to formulate AQ(H)Is for regions/countries facing different levels of air pollution – universal versus individualized guidance?

The review found that AQHIs cannot be transposed between different regions and must be specifically developed and employed for each location (e.g. country or region). In theory, universal AQIs could be developed based on international standards such as the WHO AQG levels. Nevertheless, the significant shortcomings of this approach (outlined in sections 2.2.4 and 2.5) undermine the potential merits. AQIs, like AQHIs, are best designed and employed in a country-specific manner so that they are applicable to the relevant populations. However, it is reasonable to consider also providing a context for AQ(H)Is to the public by describing the extent to which local pollutant levels exceed a global reference such as WHO AQG levels. As an example, the United Nations Environment Programme's global monitoring network provides $PM_{2.5}$ levels and a comparison with the annual WHO AQG level of 5 µg/m^3 *(183)*. For the purposes of formulating AQIs, which communicate short-term air quality information to the public, it may be important to also include a comparison with the 24-hour AQG level of 15 µg/m^3.

> **Consideration 8:** harmonize data for low-cost sensors (LCS) and integrate into official AQ(H)I systems, including calibration, data fusion and implications for development.

AQ(H)I values derived from instruments operated in official networks, such as those for regulatory monitoring, are accurate and precise but, because of their high cost and maintenance requirements, cannot provide a dense coverage throughout all areas of interest. Through advances in sensing technologies and data processing algorithms, LCS have emerged to complement official or government facilities. When carefully integrated into conventional equipment (e.g. regulatory monitors, satellite air quality products and/or predictive models), these devices can serve two key stakeholders: (i) regulatory bodies, by reducing the knowledge gap in air pollution concentrations for areas not covered by regulatory monitoring networks, as well as increasing the spatial resolution of monitoring in others; and (ii) citizens, by providing useful air quality information and increasing effective engagement. It is also tempting to deploy LCS where no official monitoring exists, such as in low- and middle-income countries or remote locations throughout the world. As an example, the United Nations Environment Programme's global monitoring network crowdsources data from a variety of sources, including LCS and regulatory monitors *(183)*. However, data from these independently operated LCS need to be thoroughly scrutinized, which poses considerable challenge for resource-limited operators.

Gaps remain in understanding the reliability of LCS and their optimal use under real-world conditions (e.g. variable temperature and humidity, locations where the important emission sources differ from those at the sensor calibration site). Therefore, once installed, regular calibration and maintenance of LCS through ongoing reference to relatively nearby high-accuracy government regulatory monitors (e.g. supersites), as well as careful data quality assurance checks, is essential. For example, in the Breathe London Communities Network, community-owned LCS are mathematically linked to regulatory monitors to improve data accuracy *(39)*.

In addition to monitoring in urban environments, the use of blended AQ(H)I data by combining output (i.e. crowdsourcing) from LCS, regulatory monitors and air quality models has potential to accurately and responsively address monitoring challenges in areas where regulatory monitors are

sparsely distributed (e.g. during widespread episodic events such as wildfire and dust storm incidents). However, even in such situations, the capability of LCS to measure $PM_{2.5}$ has received the most attention and acceptance, for example, to augment government monitoring data and in the Unites States' AirNow Fire and Smoke Map *(185)*. Nonetheless, LCS data are still considered less accurate and are best used in conjunction with the official network data. LCS have more difficulty with the stable, accurate, interference-free measurement of important trace gases such as NO_2 and O_3 or of larger particles (i.e. PM_{10}). In summary, there is growing potential for the use and integration of LCS into monitoring networks to enhance the resolution and breadth of AQ(H)I coverage. However, several key barriers and unresolved issues need to be overcome for this approach to evolve into mainstream policy.

3.2 Communication

> **Consideration 9:** optimize risk categorization systems for public communication such as derived numerical scores versus alternative strategies (e.g. providing pollution concentrations).

While it is practical and useful to issue a single overall AQ(H)I value as a reasonable strategy to enable clear and concise public messaging (as stated in Consideration 3), this process lacks full transparency of information. The concentration (or AQI level, when relevant) of each individual pollutant measured at each location should also be accessible (as discussed in Consideration 2). This straightforward modification could be of significant benefit for individuals susceptible to specific pollutants (e.g. people with asthma, who are sensitive to high O_3 levels).

> **Consideration 10:** optimize risk communication language/messaging in AQ(H)Is to accurately/effectively convey health risks to the public.

Ensuring effective communication strategies for AQ(H)Is is of pivotal importance. Regardless of the merits of other factors (e.g. the sophistication of monitoring and exposure modelling), AQ(H)Is cannot effectively serve to protect public health without successful public communication. Communicating the health risks of air pollution faces multiple barriers, including technical, impersonal and/or ambiguous language that fails to convey personal health relevance or urgency. A one-size-fits-all approach overlooks cultural norms, education, language and community-specific needs, while also failing to address vulnerable groups.

AQ(H)Is should be designed to incorporate terminology that clearly links pollutant levels to potential health impacts, especially for vulnerable groups and susceptible populations. Framing health risks in personally relevant terms and using clear, relatable language are essential. Providing near-real-time, localized air-quality data from trusted sources also facilitates their translation to tightly coupled actions. Messages should integrate negative cues about poor air quality and health effects with positive reinforcement for behaviour change, while recognizing that capability, opportunity and motivation need to align to achieve a lasting impact.

Consideration 11: enhance AQ(H)I communication strategies and modalities to improve public awareness of and potential adherence to behavioural guidance.

Enhancing public awareness of the adverse health effects of air pollution to improve awareness of and adherence to guidance is of central importance. It requires shifting from top-down messaging to co-produced, community-driven approaches. Leveraging mass media, smartphone technology and social media can expand reach but must be coupled to safeguards against misinformation and efforts to maintain institutional responsibility.

Interactive participatory research methods (e.g. involving residents in air quality education and joint policy review with statutory bodies) help to uncover local priorities and contextual knowledge that make interventions more relevant and trusted. Providing practical, culturally tailored and actionable information on reducing both exposure and emissions builds self-efficacy and strengthens the belief that individual and collective action can make a difference, thus reducing the perception that air pollution exposure cannot be influenced at personal level.

Communication should present a menu of possible behaviours (rather than a single prescribed action) by combining protective strategies (e.g. avoiding hot spots) with proactive co-benefit options such as walking or cycling instead of driving, which simultaneously improves health, reduces costs and cuts emissions. Including supportive behaviours such as advocacy and community organizing can further create enabling environments for sustained policy change by reinforcing the link between personal action and the broader societal impact.

3.3 Validation

Consideration 12: how to optimally validate the accuracy of AQ(H)I (local/personal level).

Optimal validation of the accuracy of AQ(H)Is lies within the framework for evaluating their overall merits, identifying specific limitations and helping to elucidate potential pathways for refinement (outlined in section 2.3). Specifically, this requires an assessment of their accuracy in (i) predicting local air pollution exposures and (ii) using local health data to evaluate the associations between AQ(H)I strata with health risks. If these elements are not adhered to, then behaviour modification in response to information reported through an AQ(H)I may not result in the desired reductions in personal exposure or improvements in public health. AQ(H)Is must also be validated by season because ambient pollution mixtures and interactions with climatic elements (e.g. heat) change over the course of the year.

> **Consideration 13:** how to evaluate the benefits (exposure reduction, health outcomes, low-cost monitoring versus personal-level monitoring) of following activity guidance; and how to assess the overall public health effectiveness (e.g. population-level risks/benefits) of health-based AQI.

The optimal evaluation of behaviour changes in relevant populations and measurable health benefits also lie within the framework for evaluating the overall merits of AQ(H)I systems, identifying specific limitations and helping to elucidate potential pathways for refinement (outlined in section 2.3). There is evidence that individuals, especially those with chronic diseases (e.g. asthma) and older adults, alter their behaviour in response to AQ(H)I guidance or air quality alerts. However, there is also a paucity of data in this area. Therefore, additional research is needed to understand how individuals use AQ(H)Is: whether they consult it regularly, whether they change behaviour in response to information from the AQ(H)I, and what specific actions they take in response to the index values and associated health messages.

Data are also lacking on the most important, and at the same time challenging, factor when evaluating AQIs – public health effectiveness. The available evidence shows limited public health benefits from issuing AQ(H)Is in WHO Region of the Americas and WHO European Region and evidence of some health benefits related to issuing active air quality alerts in WHO Western Pacific Region. This is a challenging area for research. No single approach is most valid or robust, given that randomized controlled trials cannot be viably performed. Study methods and statistical analyses will need to be suitable for the type and unique aspects/limitations of the data being evaluated. Many of the approaches used in studies to analyse observational data (outlined in Table 2), such as a regression discontinuity design, can provide important insights. This field of science will most benefit by accruing evidence derived from multiple differing study types and analytical methods.

> **Consideration 14:** Evaluate the level of awareness of AQ(H)I systems and its impact on behaviour and health outcomes.

Awareness of AQ(H)I systems is often cited as a benefit but is rarely measured. However, it is important to identify and address misconceptions in less-aware population groups, as well as in informed or more proactive citizens. More accurate measures of awareness may be achieved by focusing on susceptible populations beyond people with asthma or those with CVD (e.g. older people, pregnant women, overweight individuals, people with diabetes, and those who exercise close to traffic), assessing differences in the perceptions of key stakeholders (e.g. doctors, professional drivers, government officials, industry, activists, journalists, recreational runners) and conducting research into the extent to which scientists (rather than government alone) are involved in educating the public. Use of a more-standardized metric for public awareness would ensure that the results of studies evaluating awareness are more comparable. Awareness of AQ(H)I systems cannot be guaranteed to support a positive change in behaviour and health outcomes since people are not passive receptors of scientific information and may question or dismiss it. Furthermore, receiving such information may be associated with increased anxiety/stress/fear/panic, which could add to the burden of disease. Lastly, while the issuance of relatively real-time AQ(H)I values or air quality alerts is important, it is equally critical to incorporate predicted (or forecasted) values for subsequent days into the system so that people can adequately plan their activities to prevent exposures.

3.4 Further refinement and global equity

> **Consideration 15:** how/when to refine strategies as scientific knowledge advances (e.g. health risks of pollutants) and pollutant levels and baseline populations risks change over time.

The need to continually evaluate and refine the performance of AQ(H)Is is critical to ensure that the time, effort and cost involved in development and maintenance translate into responsiveness and applicability. Efforts at continuous improvement should focus on the emerging scientific evidence; clinical advice and management; ageing population/future changes in the underlying health status of the population; adaptive policies, actions and technologies (e.g. emission, net-zero); and public engagement (e.g. big data analytics, communication channels for data dissemination). Examples of relevant emerging scientific evidence may include the health risks associated with a shifting air pollution mixture (e.g. non-exhaust emissions, wildfire smoke, secondary inorganic aerosol, biogenic emissions, aeroallergens) under climate change, combined effects with other environmental stressors (e.g. heat, noise, pollen) and identification of additional air pollutants for inclusion.

> **Consideration 16:** potential for health-based AQIs for longer-term (weeks, years) pollution levels and refining AQIs based on chronic background levels.

AQ(H)Is aim to inform and protect the public about short-term elevations in air pollution. However, the largest proportion of adverse public health effects is due to long-term exposures, even at low pollution concentrations below thresholds for action in AQ(H)Is. Steps towards informing the public and raising awareness about chronic air pollution levels and related health risks should be considered, as reviewed in the working paper, *Long-term air quality statement (186)*. Long-term (e.g. annual) air pollution averages, as well as their country-specific (and, perhaps, global) relative ranking, should be communicated as part of AQHI systems (using a tiered information approach (described in section 2.1.3). Straightforward access to these data accords with the general principle of full transparency of information. It could also provide a valuable service to the public, especially for susceptible populations, such as by helping to inform individuals or families about the long-term health risks from air pollution in the areas where they reside, work, go to school or travel. However, compared with taking relatively brief actions to avoid exposures to pollution over a few hours to days, modifying life activities to reduce chronic exposures may be a more difficult or burdensome task (e.g. changing residence, altering daily commutes or outdoor activities).

> **Consideration 17:** approaches to improve the effectiveness and risk characterization/activity guidance of AQ(H)Is to protect public health.

Current AQ(H)Is provide behavioural guidance for the general (healthy) public. Sensitive individuals (e.g. older adults, children, people with pulmonary disease or CVD) are typically advised to follow the same guidance but at lower AQI/AQHI thresholds. This general approach is logical and helps to align activity advice with susceptibility to health effects. Even so, many thousands to millions of individuals must follow behavioural advice on any given day in order to prevent a single adverse health event. Emerging evidence shows that the concept of personalized or precision risk messaging

may be relevant for AQ(H)Is. This approach could reduce the number of individuals required to follow behavioural advice by better tailoring the intensity of advice to more precise estimates of underlying health risks across the population – especially for individuals without known pulmonary disease or CVD. There is increasing wide-scale availability of methods to relatively easily calculate CVD or pulmonary risk (e.g. online tools, smartphone apps). However, more research is needed to determine whether adopting a more personalized approach is an effective and viable strategy to improve future AQ(H)Is.

> **Consideration 18:** develop/validate and communicate/message AQ(H)Is, especially in countries with limited resources and limited air pollution monitoring.

Given that AQ(H)Is cannot simply be transplanted from other regions, there is no feasible alternative to the development of locally derived and adapted AQHIs. Low-cost monitoring networks and simplified risk scales designed for local literacy and cultural contexts are critical. Pairing data from LCS with mobile phone alerts, community radio and social media can extend their reach even in areas with limited infrastructure. Crucially, communication should empower citizens, not merely warn them. Participatory air quality measurements to allow citizens to appreciate the links between poor air quality and respiratory problems, missed work or school days, and quality of life and, conversely, the benefits of clean air allow communities to appreciate the benefits of lowering pollution. Partnerships with local leaders, schools and health workers can embed these messages into trusted communication channels. Over time, such locally tailored, participatory approaches can not only improve awareness and self-protection but also drive community advocacy for cleaner air.

> **Consideration 19:** incorporate air quality into health care – education, guidelines, risk estimators and electronic medical records.

The risks posed by environmental exposures need to be better incorporated into health-care practice. This involves better education and ongoing training for health-care professionals on the effects of air pollution on health and how to advise their patients. In recognition of this unmet need, WHO has launched an educational programme for health care workers *(31)*. An increasing number of global medical societies also formally recognize the importance of air quality (e.g. American Heart Association, American Lung Association, American Thoracic Society, Canadian Association of Physicians for the Environment, European Respiratory Society, European Society of Cardiology, Indian Medical Association, Royal College of Physicians (United Kingdom) and World Heart Federation). There may also be value in directly incorporating AQ(H)I data into electronic health/medical record systems. Having this information readily available for patients and health-care providers could help to improve care for susceptible patients, such as people with pulmonary disease (e.g. asthma) or CVD, who are sensitive to air pollution. Lastly, studies have shown that information related to air pollution exposure (i.e. the "envirome") provides incremental risk prediction for CVD and other diseases. It is plausible that risk calculators used in clinical practice could incorporate environmental determinants of risk, including air quality, in the future.

> **Consideration 20:** in addition to warning individuals about the health risk(s) of daily pollution levels, what actions can/should governments be taking to reduce risks on such occasions?

The most direct way to potentially reduce population exposures during periods of elevated AQ(H)I levels, which are often due to certain weather patterns, is through widespread public action to reduce pollutant emissions. Multiple cities/countries initiate restrictions on the number or types of vehicles permitted on roads during these periods (e.g. only odd or even license plate numbers, no high-emitting diesel vehicles) or restrictions on the use of certain roads (e.g. greater restrictions on entry into congestion-charging or clean air zones) or vehicle speeds. Some jurisdictions also use incentives not to drive, such as free/reduced-fare mass transit use. In some countries, high-emitting industries (e.g. steel, cement) and/or fossil-fuel-based power plants are required to reduce their output or cease operations. Construction/demolition projects, which are significant sources of dust, may be required to reduce activity and/or increase dust suppression measures. Use of solid fuels for heating (e.g. coal) may also be restricted during high AQ(H)I periods. The efficacy of such interventions has been evaluated and improvements in air quality have been demonstrated. These actions in general are needed; however, since compliance is critical, enforcement measures may be necessary to realize the full intended benefit.

> **Consideration 21:** incorporate other environmental exposures (e.g. temperature, noise, pollen) in the context of climate change.

Given the additive acute health effects of other environmental exposures, and an increased likelihood of some such interactions because of climate change (including non-optimal daily temperatures, pollen and environmental noise levels), moving towards including estimates of these risks and actionable advice into AQ(H)I systems may help to protect public health. Incorporating multiple exposures into a single multi-exposure composite index is an aspiration, but is complicated by the fact that each exposure poses different health impacts, for example, pollen increases pulmonary events, heat worsens mortality and cold affects cardiovascular events.

4. Conclusions

Based on the comprehensive review of the scientific literature, deliberations of the expert writing group, and external review, consensus was reached on many issues related to roadmaps for advancing the development, communication, validation, and further refinement and global equity of AQ(H)Is. Considerations and comments are summarized as follows.

Development

- Provide practical guidance to the public on how to reduce exposures as a component of AQ(H)Is.
- Consider transitioning from a conventional AQI to a country- or location-specific multipollutant AQHI, wherever possible.
- Communicate a single, unitless value as an overall summary AQ(H)I.
- Consider strategies for communicating the actual health risk estimates associated with daily AQHI values to the public (i.e. improved transparency of information).
- Consider providing the concentrations, or AQI values, of all individual measured air pollutants (i.e. improved transparency of information).
- Use AQ(H)I categories to facilitate the communication of air quality and activity advice to the public.
- Most AQ(H)Is currently promote reasonable behavioural advice choices.
- The benefits and potential risks/costs of advocating for more aggressive protective actions remain unclear (e.g. face coverings, air cleaners).
- Formulate improved (e.g. more accurate or comprehensive) multipollutant AQHIs;
- Develop a $PM_{2.5}$-AQHI as a reasonable interim/alternative option to formulating a full multipollutant AQHI.
- When using conventional AQIs, consider renaming any misleading categories (e.g. good air quality) to reflect that health risks exist at low exposure levels.
- Consider developing and communicating subcategories of disease-specific AQHIs.
- Country-specific AQHIs or communication thresholds based on a common risk estimation model are warranted because these cannot be transposed among locations owing to large differences in concentration levels and stakeholder perceptions between countries.
- Consider providing a global context for reported AQ(H)Is (e.g. comparison to WHO AQG levels).
- Consider involving LCS in AQ(H)I calculations, when validated and useful.

Communication

- Consider increasing public transparency for all measured air pollution/health risk data.
- Consider a tiered information approach for communicating complex/additional data.
- Consider improved/culturally aware approaches for AQ(H)I communication.
- Consider new approaches to improve AQ(H)I awareness/adherence.
- Consider greater choices in activity guidance to overcome barriers for specific communities.

Validation

- Use a tiered approach for validating and evaluating AQ(H)Is.
- Use multiple study types to evaluate the effectiveness of AQ(H)Is.
- Several new approaches to evaluate AQ(H)I adherence are reasonable.

Further refinement and global equity

- Evaluate, refine and update AQ(H)Is on an ongoing basis.
- Consider developing and communicating AQ(H)Is related to long-term exposures.
- Consider providing refined activity advice tailored to more personalized risk estimates.
- Strategies to improve communication in resource-limited areas are reasonable.
- Consider adopting AQ(H)I information and education into clinical medical practice.
- There are reasonable actions that governments can take to reduce emissions during alerts.
- Consider adopting additional environmental exposures into composite indexes.

Key research priorities are to:

- conduct research to identify novel approaches to evaluate and quantify the public health benefits (and potential risks) of issuing AQ(H)Is and air quality alerts across multiple populations and the global spectrum of air quality;
- develop/validate strategies for effectively communicating the health risk estimates associated with AQHI values to the public; and assess the benefits (e.g. improved adherence or health outcomes) and pitfalls;
- assess the benefits (e.g. improved adherence) and pitfalls (e.g. increased complexity, poor public understanding) of communicating the concentrations of all measured air pollutants;
- assess the comparative effectiveness (i.e. exposure reductions, health benefits) of following specific AQ(H)I activity changes as advocated by current guidance, as well as more aggressive actions (e.g. use of face coverings, air cleaners);
- determine the public health benefits/risks of additional strategies to enhance future AQ(H)Is, such as providing subsets of disease-specific AQHIs, issuing more-personalized activity advice tailored to health risks, and providing comparisons to WHO AQG levels and information/advice on location-specific long-term air quality;
- improve the validation and use of LCS and other methods to enhance monitoring networks and exposure modelling for formulating AQ(H)Is, especially in resource-limited locations;
- assess the public health benefits of strategies to incorporate AQ(H)I information into clinical medical practice, including electronic medical records, medical education and clinical guidelines; and
- assess the public health benefits of strategies to incorporate additional environmental exposure information (e.g. heat, pollen) into a composite index.

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